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Stability and instabilities of 2D Euler equations on the torus

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Abstract

This project aims to present the main results of the paper *Instability of Two-Dimensional Taylor–Green Vortices* by Cao-Labora, Colombo, Dolce, and Ventura [3]within a didactic framework accessible to Master's students in mathematics. In the final part, we presents an original contribution on the instability of (2,2) Taylor-Green vortex developed during the semester project, building on the pioneering method introduced in the reference paper.

1 Introduction

The motion of a two-dimensional incompressible ideal fluid evolving in a periodic box is described by the classical Euler equations. In vorticity formulation, they can be written as

$$\begin{cases} \partial_t \omega + u \cdot \nabla \omega = 0 & \text{for } (x, y) \in \mathbb{T}^2,\ t \geq 0, \\ u = \nabla^\perp \psi, & \\ \Delta \psi = \omega. & \end{cases} \tag{1.1}$$

A central problem in the mathematical study of the Euler equations is the spectral stability of steady solutions. For some classes of stationary flows, such as shear flows (e.g. Couette or Poiseuille) and radial vortices, the spectral picture is by now well understood in a classical line of literature starting from the works of Rayleigh, Kelvin and Tollmien [10, 13, 14]. An important contribution was made by Arnold, who introduced useful methods based on the Hamiltonian structure of the two-dimensional Euler equations [1, 2]. By contrast, for more general steady states with non-unidirectional streamline geometry, only a handful of cases have been studied, such as certain cat's-eye flows [7] and Kelvin–Stuart vortices [8], and many fundamental questions still remain open.

Taylor–Green vortices belong to this second class of steady states with genuinely multidimensional geometries. Originally, they were introduced in a 3D formulation by Taylor and Green to study the generation of small eddies from large scale structures [12]. In the 2D setting considered throughout this report, Taylor–Green vortices provide a classical and analytically explicit family of steady solutions on the two-dimensional torus, and they also play an important role as benchmark configurations in fluid dynamics [11]. For this reason, they offer a natural testing ground for new methods in spectral stability theory. The recent paper by Cao-Labora, Colombo, Dolce, and Ventura develops a new instability criterion for linear Hamiltonian operators and applies it to the study of the Taylor–Green vortex [3].

The aim of this report is twofold. First, we present some of the main ideas and results of the original paper in a form intended to be accessible to Master's students in mathematic. Second, building on the theoretical framework introduced there, we discuss a further case of Taylor-Green vortex not treated in the original article and present a new result obtained during the semester project.

1.1 Historical background. Navier–Stokes equations are one of the most famous partial differential equations of our time. The matter of existence of solutions for these equations is still widely regarded as one of mathematics most pressing open problems. This is reflected in the fact that the three-dimensional existence and smoothness problem is included among the *Millennium Prize problems*: a series of seven open problems presented in the year 2000 by the Clay mathematics institute, with a prize of 1.000.000$ assigned to each of them [4]. In its official formulation, the problem asks for a proof either of global existence and smoothness of physically reasonable solutions, or of finite-time breakdown, for the incompressible Navier–Stokes equations in three space dimensions [5].

Navier–Stokes equations are a system of nonlinear partial differential equations relating velocity and pressure derived from elementary conservation principles and Newtonian laws of motion.
They constitute the most general mathematical description of fluid motion, and they have proved beyond successful in describing a large variety of fluid behavior in real world applications. We refer to [9] for a more detailed discussion of Navier–Stokes and their relevance.

The incompressible Navier–Stokes equations for $u = u(t, x)$ and $p = p(t, x)$ on a domain Ω and forcing term f read as follows

$$\begin{cases} \partial_t u + (u \cdot \nabla) u = -\nabla p + \nu \Delta u + f, \text{ in } \Omega & \text{(conservation of momentum)}, \\ \nabla \cdot u = 0, \text{ in } \Omega & \text{(conservation of mass)}. \end{cases} \tag{1.2}$$

In a nutshell, the first equation of system (1.2) relates the derivative of momentum on the left hand side, with the force per unit mass acting on the fluid, on the right hand side. Different force contributions can be identified based on their nature. $-\nabla p$ represents surface forces due to the difference in pressure, $\nu\Delta u$ is the force between fluid particles due to viscosity and f is an external forcing term per unit mass, e.g. gravity.

2 2D Euler equations on the torus

In this report we will study the two-dimensional Euler equations in vorticity formulation on a torus. We recall only the basic derivation suitable to our purposes; for a complete discussion on vorticity methods and incompressible flows we refer to Majda and Bertozzi [9]. The Euler equations can be obtained from the Navier–Stokes in (1.2) by assuming that viscous effects are absent. Fluids satisfying this set of assumptions are usually referred to as *perfect fluids*.
In the inviscid case without external forcing, namely when $\nu = 0$ and $f = 0$, the N-S system (1.2) reduces to the incompressible Euler equations:

$$\begin{cases} \partial_t u + (u\cdot\nabla)u + \nabla p = 0, \text{ in } \Omega & \text{(momentum equation)},\\ \nabla\cdot u = 0, \text{ in } \Omega & \text{(incompressibility constraint)}. \end{cases} \tag{2.1}$$

In order to pass to the formulation used throughout this report, two further modifications are required. First, in two dimensions the incompressibility condition $\nabla\cdot u = 0$ allows to introduce a stream function ψ related to our velocity field as follows

$$u = \nabla^{\perp}\psi, \qquad \nabla^{\perp} := (-\partial_y, \partial_x).$$

With this convention, the scalar vorticity is defined by

$$\begin{aligned} \omega : &= \mathrm{curl}(\ u\ ) = \nabla^{\perp}\cdot u = \partial_x u_2 - \partial_y u_1,\\ &= \nabla^{\perp}\cdot\nabla^{\perp}\psi = \Delta\psi. \end{aligned}$$

Second, to obtain the vorticity formulation on the torus we need to take the curl of the equations and work with ω as our unknown. Taking the curl of the momentum equation in (2.1) removes the pressure term, since $\mathrm{curl}(\nabla p) = 0$. Moreover, a standard computation in the $2D$ case, shows that $\mathrm{Curl}((u\cdot\nabla)u) = u\cdot\nabla\omega$. Therefore the momentum equation in system (2.1) can be rewritten in vorticity formulation as $\partial_t\omega + u\cdot\nabla\omega = 0$.
From this point on, we work on the periodic domain $\mathbb{T}^2 := \mathbb{R}^2/(2\pi\mathbb{Z}^2)$, so that $(x, y) \in \mathbb{T}^2, \quad t \geq 0$. Employing a toroidal domain means that all functions are identified under translation by integer multiples of 2π in each spatial direction. In other words, the unknowns are 2π-periodic in both x and y:

$$u(x + 2\pi k_1, y + 2\pi k_2, t) = u(x, y, t), \qquad (k_1, k_2) \in \mathbb{Z}^2.$$

We can finally write the two–dimensional Euler equations in vorticity formulation on the torus. They constitute the main object of study through this report.

$$\begin{cases} \partial_t\omega + u\cdot\nabla\omega = 0, \text{ with } (x, y) \in \mathbb{T}^2,\ t \geq 0,\\ u = \nabla^{\perp}\psi,\\ \Delta\psi = \omega. \end{cases} \tag{2.2}$$

2.1 Choice of a suitable basis. As a preliminary step, it is convenient to introduce a basis for the underlying function space which is naturally adapted to the periodic geometry of the problem and to the action of the operator. Since the equations are studied on the 2D torus $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z}^2)$, we work with functions that are 2π-periodic in each variable. This periodicity forces the admissible Fourier modes to have integer frequencies, so the natural basis is given by the complex exponentials

$$e_{j,k}(x, y) := e^{\,i\,(jx+ky)}, \qquad (j, k) \in \mathbb{Z}^2.$$

Indeed, the condition $e^{i(j(x+2\pi)+ky)} = e^{i(jx+ky)}$ and $e^{i(jx+k(y+2\pi))} = e^{i(jx+ky)}$ holds exactly when $j, k \in \mathbb{Z}$.
However, since the linearized operator involves the inverse Laplacian Δ^{-1}, we must restrict ourselves to a space on which this operation is well defined. On $\mathbb{T}^2$, the kernel of the Laplacian contains all constant functions, because $\Delta C = 0,\ \forall C \in \mathbb{R}$. Therefore, in order to define Δ^{-1} uniquely, we restrict ourselves to functions with zero average, namely

$$L^2_0(\mathbb{T}^2) := \left\{ f \in L^2(\mathbb{T}^2; \mathbb{C}) \ :\ \int_{\mathbb{T}^2} f(x, y)\, dx\, dy = 0 \right\}. \tag{2.3}$$

On this space, the constant Fourier mode is removed, and the inverse Laplacian is defined by inverting the Laplacian on each non-zero Fourier mode. In the following we shall use standard facts on Hilbert spaces, orthogonal decompositions and orthogonal projections [6].

To obtain a Hilbert space, we endow $L^2_0(\mathbb{T}^2)$ with the usual L^2 complex scalar product

$$\langle f, g\rangle_{L^2} := \int_{\mathbb{T}^2} f(x,y)\,\overline{g(x,y)}\,dx\,dy, \tag{2.4}$$

which defines the following induced norm

$$\|f\|_{L^2} := \left(\int_{\mathbb{T}^2} |f(x,y)|^2\,dx\,dy\right)^{1/2}. \tag{2.5}$$

3 Linearization around a steady Euler equilibrium

We are often interested in understanding how deviations from an equilibrium affect the dynamics of the solution. Since perturbations start initially close, in a suitable norm, to the reference steady state, a natural and powerful tool to understand their evolution is the linearized operator.
In principle one could linearize the equation in any point of the phase space (where, instead of point, it wouldn't be wrong to read 'function'), however, the most relevant case is doing so around steady state solutions. These are time independent solutions of the equations which maintain the same shape at all times.

Before presenting the linearized Euler operator, it is useful to define some useful notation.
In their general form, *Taylor–Green vortices* have the following analytical expression

$$\psi_{m,n} = -\sin(mx)\sin(ny),\ \ m,n \in \mathbb{N} \tag{3.1}$$

and they are all steady states for the Euler operator.
Indeed $\omega_{m,n} = \Delta\psi_{m,n} = -(m^2+n^2)\psi_{m,n}$, and $u_{m,n}\cdot\nabla\omega_{m,n} = -(m^2+n^2)\,u_{m,n}\cdot\nabla\psi_{m,n} = -(m^2+n^2)\,\nabla^\perp\psi_{m,n}\cdot\nabla\psi_{m,n} = 0$, thus

$$\partial_t\omega_{m,n} = -u_{m,n}\cdot\nabla\omega_{m,n} = 0,$$

In the following we will restrict our attention to the (1,1) Taylor–Green equilibrium, simply referred to as Taylor–Green vortex. Its vorticity, stream function and velocity are respectively given by

$$\omega_E(x,y) := 2\sin x\sin y, \qquad \psi_E(x,y) := -\sin x\sin y, \qquad u_E := \nabla^\perp\psi_E. \tag{3.2}$$

Similarly to the general Taylor–Green vortices, it satisfies the following relation $\omega_E = F(\psi_E)$ with $F(s) = -2s$.

It is also convenient to introduce at this moment the Poisson bracket notation

$$\{f,g\} := \nabla^\perp f\cdot\nabla g = \partial_x f\,\partial_y g - \partial_y f\,\partial_x g.$$

Note that $\{f,f\} = 0$ since $\nabla^\perp f$ is obtained by rotating ∇f by 90° clockwise and is therefore orthogonal to the same vector without rotation.

We are now in a position to linearize equation (2.2) around the equilibrium point ω_E (3.2). This will yield the desired linear operator $\mathcal{L}_E : L^2_0(\mathbb{T}^2) \to L^2_0(\mathbb{T}^2)$. To define it, we will characterize its action on an arbitrary function w. This is achieved by perturbing the steady state ω_E in the direction of w and setting the value of $\mathcal{L}_E\,w$ as the derivative of the original operator acting on $\omega_E + \varepsilon\,w$ evaluated at $\varepsilon = 0$.

First, we write $\omega = \omega_E + \varepsilon w$ and correspondingly $\psi = \psi_E + \varepsilon\Delta^{-1}w$, where ε is a small parameter. Substituting this ansatz into (2.2), we obtain

$$\partial_t(\omega_E + \varepsilon w) + \{\psi_E + \varepsilon\Delta^{-1}w, \omega_E + \varepsilon w\} = 0.$$

$$\partial_t(\omega_E + \varepsilon w) + \{\psi_E, \omega_E\} + \varepsilon\{\psi_E, w\} + \varepsilon\{\Delta^{-1}w, \omega_E\} + \varepsilon^2\{\Delta^{-1}w, w\} = 0.$$

Using that ω_E is a steady state, namely $\partial_t\,\omega_E = -\{\psi_E, \omega_E\} = 0$ yields

$$\varepsilon\,\partial_t w + \varepsilon\ \{\psi_E, w\} + \varepsilon\ \{\Delta^{-1}w, \omega_E\} + \varepsilon^2\{\Delta^{-1}w, w\} = 0.$$

Finally, by deriving with respect to ε in the point $\varepsilon = 0$, we obtain

$$\partial_t\,w = \mathcal{L}_E$$
$$\mathcal{L}_E(w) := -\{\psi_E, w\} - \{\Delta^{-1}w, \omega_E\}.$$

Using $\omega_E = F(\psi_E)$ and the antisymmetry of the Poisson bracket, we can rewrite the second term as $\{\Delta^{-1}w, \omega_E\} = \{\Delta^{-1}w, F(\psi_E)\} = -2\{\Delta^{-1}w, \psi_E\} = 2\{\psi_E, \Delta^{-1}w\}$. Therefore the final linearized equation reads $\partial_t w = \mathcal{L}_E(w)$, where

$$\mathcal{L}_E(w) = -\{\psi_E, (\mathrm{Id} + 2\Delta^{-1})w\}. \tag{3.3}$$

This is the linearized Euler operator around the Taylor–Green vortex, and its spectral properties will be the main object of study in the rest of the report.

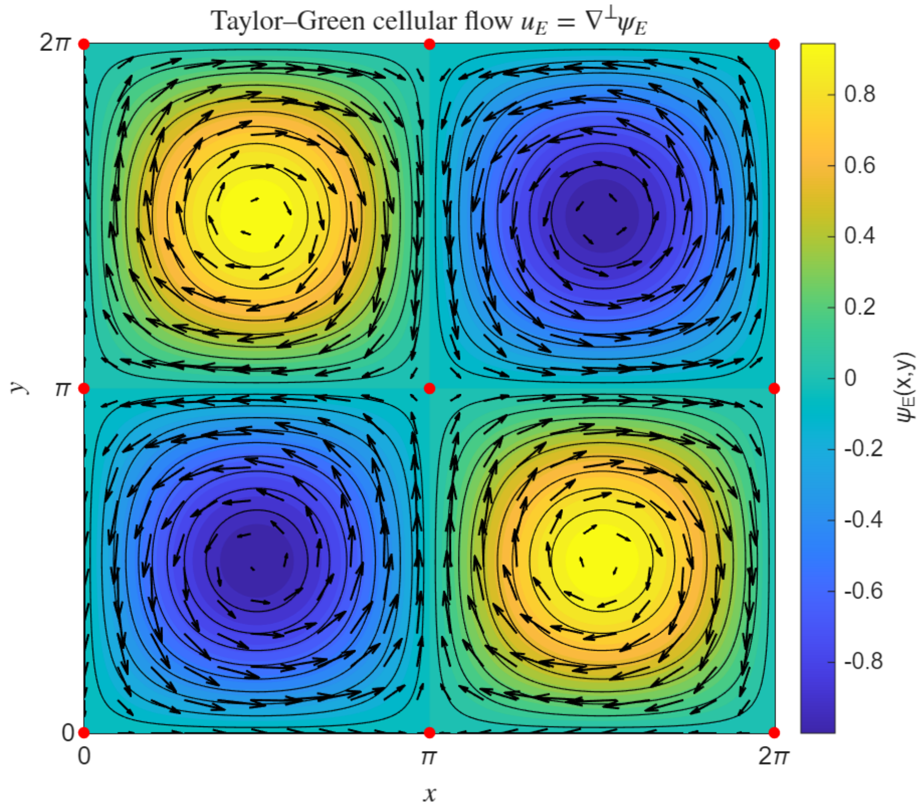

Figure 1: Velocity field and streamlines of the $(1, 1)$ Taylor–Green cellular flow on the periodic cell $[0, 2\pi]^2$. The arrows represent the velocity field $u_E = \nabla^{\perp}\psi_E$, while the background shows level sets of the stream function $\psi_E(x, y) = -\sin x \sin y$. The plot highlights the characteristic cellular structure of the equilibrium flow.

4 Hamiltonian operators

The purpose of this section is to identify some structural properties of the linearized Euler operator that will later justify its decomposition into two parts with qualitatively different behaviour. This point of view is classical in the study of the Euler equations and provides the natural framework for the spectral analysis carried out in the rest of this report.

All the results and proofs in Sections 4 and 5 are presented in a general form, valid for any *Hamiltonian operator*. This level of generality is useful, as it will allow us to apply the same criterion also to suitable modifications of the original operator $\mathcal{L}_E$. At the same time, we provide some intuition for the general properties and explain how they relate to our specific setting.

Definition 4.1 (Hamiltonian operator)**.** Let X be a Hilbert space. We say that an operator $\mathcal{L} = \mathcal{J}\mathcal{H}$ densely defined on X is *Hamiltonian* if the following properties hold:

1. the operator $\mathcal{J} : D(\mathcal{J}) \subset X \to X$ is densely defined, closed, and skew-adjoint;
2. the operator $\mathcal{H} : X \to X$ is bounded and self-adjoint.

4.1 Hamiltonian structure of the Taylor–Green operator. In the Taylor–Green case, the Hilbert space X becomes $L^2_0(\mathbb{T}^2)$, the scalar product $\langle \cdot, \cdot \rangle_X$ is just the standard complex scalar product in L^2 and the abstract decomposition $\mathcal{L} = \mathcal{J}\mathcal{H}$ becomes completely explicit. Indeed,

$$\mathcal{J} = -\{\psi_E, \cdot\} \quad \text{and} \quad \mathcal{H} = (\mathrm{Id} + 2\Delta^{-1}).$$

This is useful because the two operators play very different roles. The operator $\mathcal{J}$ is the transport part: it differentiates along the streamlines of the equilibrium flow. By contrast, $\mathcal{H}$ is much simpler and diagonalises in our Fourier basis. Indeed, $\Delta e_{j,k} = -(j^2 + k^2)e_{j,k}$ and $\Delta^{-1} e_{j,k} = -\frac{e_{j,k}}{(j^2+k^2)}$. Therefore $\mathcal{H}$ acts diagonally

$$\mathcal{H} e_{j,k} = \mu_{j,k} e_{j,k}\,, \qquad \text{where } \mu_{j,k} := 1 - \frac{2}{j^2 + k^2}, \qquad (j,k) \neq (0,0),$$

$$\text{equivalently } \mathcal{H} = \sum_{(j,k)\neq(0,0)} \mu_{j,k}\, e_{j,k} \otimes e_{j,k}\,, \quad \text{where } (e_{j,k} \otimes e_{j,k})\, f := \frac{\langle f, e_{j,k}\rangle_{L^2}}{\|e_{j,k}\|^2_{L^2}}\, e_{j,k}.$$

So the sign of $\mathcal{H}$ is only determined by $j^2 + k^2$:

- $\mathcal{H}$ is negative on the modes with $j^2 + k^2 \leq 1, \quad j,k \in \mathbb{Z}$;
- $\mathcal{H}$ vanishes on the modes with $j^2 + k^2 = 2, \quad j,k \in \mathbb{Z}$;
- $\mathcal{H}$ is positive on all remaining modes $j^2 + k^2 \geq 3, \quad j,k \in \mathbb{Z}$.

Hence the negative subspace of $\mathcal{H}$ is

$$N = \mathrm{span}\{\sin x, \sin y, \cos x, \cos y\}, \tag{4.1}$$

while its kernel is

$$K = \mathrm{span}\{\sin x \sin y, \sin x \cos y, \cos x \sin y, \cos x \cos y\}. \tag{4.2}$$

Indeed, the multiplier $\mu_{j,k} := 1 - \frac{2}{j^2+k^2}$ vanishes when $j^2 + k^2 = 2$, while it is negative when $j^2 + k^2 = 1$. The remaining modes, namely those satisfying $j^2 + k^2 > 2$, span the positive spectral subspace of $\mathcal{H}$, which we denote by $G := \overline{\mathrm{span}\{e_{j,k} : (j,k) \neq (0,0),\ j^2 + k^2 > 2\}}$. We can now write the orthogonal decomposition

$$L^2_0(\mathbb{T}^2) = K \oplus N \oplus G.$$

It is then natural to denote by Π_0, Π_-, Π_+ the orthogonal projections onto K, N, and G, respectively; which are to be understood in the context of L^2-inner product [6]. Taken $f = \sum_{(j,k)\neq(0,0)} c_{j,k} e_{j,k}$, they are defined as

$$\Pi_0 f = \sum_{j^2+k^2=2} c_{j,k} e_{j,k}, \quad \Pi_- f = \sum_{\substack{j^2+k^2 \leq 1 \\ (j,k)\neq(0,0)}} c_{j,k} e_{j,k}, \quad \Pi_+ f = \sum_{j^2+k^2\geq 3} c_{j,k} e_{j,k}, .$$

Where the absence of the $(0,0)$ mode is motivated by the choice of a space containing only zero average functions. Consequently, every function $f \in L^2_0(\mathbb{T}^2)$ can be decomposed as $f = \Pi_+ f + \Pi_- f + \Pi_0 f$.

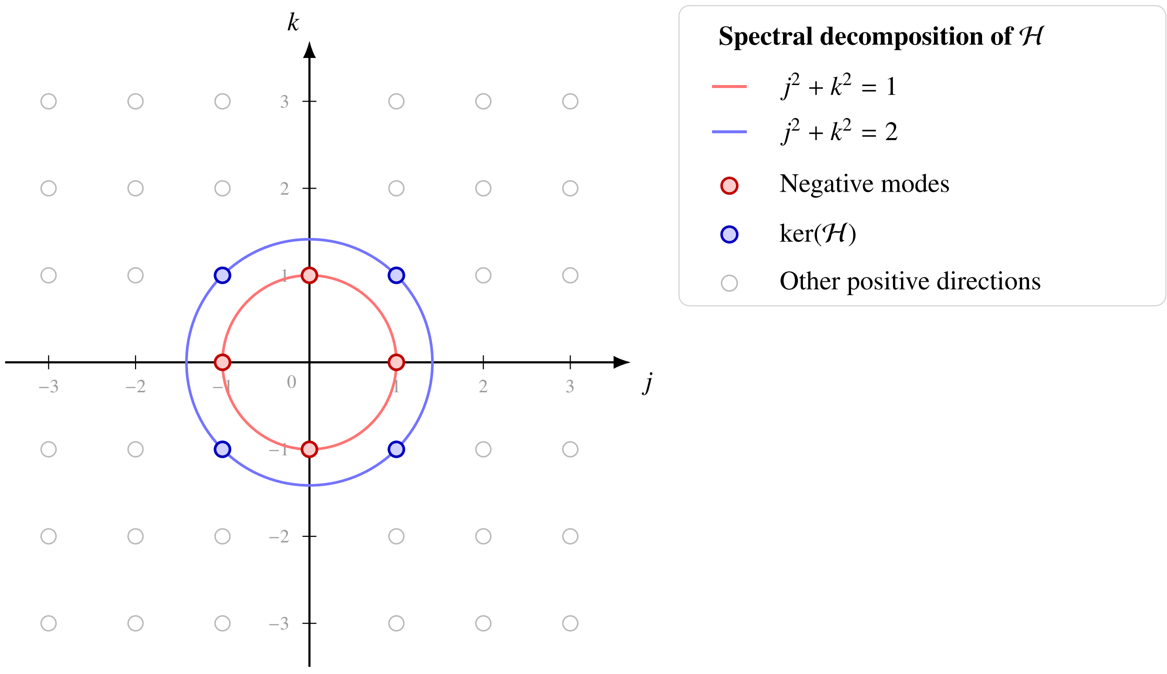

Figure 2: Illustration of the distribution of Fourier modes $(0,0) \neq (j,k) \in \mathbb{Z}^2$ in the Fourier plane. The red points denote the negative directions of $\mathcal{H}$, while the blue points denote the kernel of $\mathcal{H}$. The circles of radii 1 and $\sqrt{2}$ mark the thresholds $j^2 + k^2 = 1$ and $j^2 + k^2 = 2$.

4.2 Negative directions and instability. In this subsection we present some properties of Hamiltonian operators that will be used later in the study of the linearized Taylor–Green operator (3.3). The main idea is that unstable directions of $\mathcal{L}$ are closely related to negative directions of $\mathcal{H}$. This will be useful when we will need to decompose the dynamics into invariant subspaces and want to rule out instability in those subspaces where $\mathcal{H}$ has no negative directions.

Let $\mathcal{L} = \mathcal{J}\mathcal{H}$ be a Hamiltonian operator in the sense of Definition 4.1. Define the quadratic form

$$q_{\mathcal{H}}(f) := \langle \mathcal{H}f, f\rangle_X.$$

This quantity is conserved along the flow generated by $\mathcal{L}$. Indeed, if $f(t)$ solves $\partial_t f = \mathcal{L}f = \mathcal{J}\mathcal{H}f$, then, using that $\mathcal{H}$ is self-adjoint and $\mathcal{J}$ is skew-adjoint, we find

$$\begin{aligned}
\frac{d}{dt} q_{\mathcal{H}}(f(t)) &= \frac{d}{dt}\langle \mathcal{H}f(t), f(t)\rangle_X \\
&= \langle \mathcal{H}\partial_t f, f\rangle_X + \langle \mathcal{H}f, \partial_t f\rangle_X \\
&= \langle \mathcal{H}\mathcal{J}\mathcal{H}f, f\rangle_X + \langle \mathcal{H}f, \mathcal{J}\mathcal{H}f\rangle_X \\
&= \langle \mathcal{J}\mathcal{H}f, \mathcal{H}f\rangle_X + \langle \mathcal{H}f, \mathcal{J}\mathcal{H}f\rangle_X \\
&= 0.
\end{aligned}$$

So $q_{\mathcal{H}}$ is a conserved quantity for the linear dynamics. Note that the same proof remains true when instead of $q_{\mathcal{H}}$, we consider $\langle \mathcal{H}f, g\rangle_X$ where also g solves $\partial_t g = \mathcal{L}g = \mathcal{J}\mathcal{H}g$,.

This simple fact already gives some intuition. If f is an eigenfunction of $\mathcal{L}$, say $\mathcal{L}f = \lambda f$ with $\mathrm{Re}(\lambda) \neq 0$, then $e^{\lambda t} f$ is a solution for the evolution problem with prescribed initial condition f. Along this solution, $q_{\mathcal{H}}(e^{\lambda t} f) = e^{2\,\mathrm{Re}(\lambda)t} q_{\mathcal{H}}(f)$. Since $q_{\mathcal{H}}$ must stay constant in time, the only possibility is $q_{\mathcal{H}}(f) = \langle \mathcal{H}f, f\rangle_X = 0$. Note that this criterion gives a requirement for all eigenvectors corresponding to eigenvalues λ s.t. $\mathrm{Re}\lambda \neq 0$ not just unstable ones.
In particular, unstable eigenfunctions must lie in directions where the quadratic form of $\mathcal{H}$ vanishes. The next lemma makes this principle precise. It shows that the number of unstable directions of $\mathcal{L}$ is bounded by the number of negative directions of $\mathcal{H}$.

Lemma 4.1. *Let $\mathcal{L} = \mathcal{J}\mathcal{H}$ be a Hamiltonian operator on a Hilbert space X, such that*

1. $\mathcal{H}$ has exactly k negative eigenvalues, counted with multiplicity;

2. $\mathcal{L}$ has m linearly independent unstable eigenfunctions.

Then every unstable eigenfunction f of $\mathcal{L}$ satisfies $\langle \mathcal{H}f, f\rangle_X = 0$, and moreover $m \leq k$. In particular, if $\mathcal{H}$ has no negative eigenvalues, then $\sigma(\mathcal{L}) \subset i\mathbb{R}$.

Proof. Let $f_1, \dots, f_m$ be linearly independent eigenfunctions of $\mathcal{L}$ with eigenvalues $\lambda_1, \dots, \lambda_m$ such that $\mathrm{Re}(\lambda_j) > 0$ for every j and define

$$S := \mathrm{span}\{f_1, \dots, f_m\}.$$

We first show that the bilinear form associated with $\mathcal{H}$ vanishes on S.
Take $p, q \in \{1, \dots, m\}$. Since $\mathcal{L}f_p = \lambda_p f_p$, $\mathcal{L}f_q = \lambda_q f_q$, and using that $\mathcal{J}$ is skew-adjoint and $\mathcal{H}$ is self-adjoint, we have $\lambda_p\langle f_p, \mathcal{H}f_q\rangle_X = \langle \mathcal{J}\mathcal{H}f_p, \mathcal{H}f_q\rangle_X = -\langle \mathcal{H}f_p, \mathcal{J}\mathcal{H}f_q\rangle_X = -\bar{\lambda}_q\langle \mathcal{H}f_p, f_q\rangle_X$, thus $(\lambda_p + \bar{\lambda}_q)\langle f_p, \mathcal{H}f_q\rangle_X = 0$. Since both λ_p and λ_q have positive real part, their sum cannot be zero. Hence

$$\langle f_p, \mathcal{H}f_q\rangle_X = 0 \text{ for all } p, q \in \{1, \dots, m\}. \tag{4.3}$$

So the bilinear form $\langle \mathcal{H}\cdot, \cdot\rangle_X$ vanishes identically on S.
To justify this last step explicitly, take a generic $f \in S$, necessarily $f = \sum_{p=1}^m c_p f_p$ and therefore $\langle \mathcal{H}f, f\rangle_X = \left\langle \mathcal{H}\sum_{p=1}^m c_p f_p, \sum_{q=1}^m c_q f_q\right\rangle_X = \sum_{p,q=1}^m c_p\bar{c}_q\langle \mathcal{H}f_p, f_q\rangle_X$. Hence, by (4.3) we obtain $\langle \mathcal{H}f, f\rangle_X = 0$ for every $f \in S$.

We now prove that $m \leq k$. Since $\mathcal{H}$ is self-adjoint, the spectral theorem provides an orthogonal decomposition of the space X according to the sign of its spectrum $\sigma(\mathcal{H})$. We denote by Π_-, Π_+, Π_0 the orthogonal projections onto the negative spectral subspace, the positive spectral subspace, and the kernel of $\mathcal{H}$, respectively. In particular, the range of Π_- is the finite-dimensional subspace generated by the k negative eigenvectors of $\mathcal{H}$, counted with multiplicity, while Π_0 projects onto $\ker(\mathcal{H})$.
Our claim is equivalent to saying that the following vectors are independent

$$\Pi_- f_1, \dots, \Pi_- f_m$$

By contradiction, suppose that this is false. Then there exists a nonzero vector $f = \alpha_1 f_1 + \cdots + \alpha_m f_m \in S$ s.t. $\Pi_- f = 0$. Since $f \in S$ and the bilinear form $\langle \mathcal{H}\cdot, \cdot\rangle_X$ vanishes on S, we also have $\langle \mathcal{H}f, f\rangle_X = 0$.
Now decompose f according to the spectral splitting of $\mathcal{H}$:$f = \Pi_+ f + \Pi_0 f$, because the negative component is zero.

Now we proceed in our proof using the orthogonality of the spectral subspaces and the fact that $\mathcal{H}$ vanishes on its kernel, $\langle \mathcal{H}f, f\rangle_X = \langle \mathcal{H}\Pi_0 f, \Pi_0 f\rangle_X + \langle \mathcal{H}\Pi_+ f, \Pi_+ f\rangle_X = \langle \mathcal{H}\Pi_+ f, \Pi_+ f\rangle_X$.
At this point the key observation is that $\mathcal{H}$ is strictly positive on the positive spectral subspace. Therefore

$$\langle \mathcal{H}\Pi_+ f, \Pi_+ f\rangle_X \geq 0,$$

and equality can hold only if $\Pi_+ f = 0$. We also already knew that $\Pi_- f = 0$, it necessarily follows that

$$f = \Pi_0 f \in \ker(\mathcal{H}).$$

Finally, because $\mathcal{L} = \mathcal{J}\mathcal{H}$, every vector in $\ker(\mathcal{H})$ also belongs to $\ker(\mathcal{L})$. This is impossible: the vector f is nonzero and lies in the span of eigenfunctions whose eigenvalues have positive real part, so it cannot belong to the kernel of $\mathcal{L}$. This contradiction proves that $\Pi_- f_1, \dots, \Pi_- f_m$ are linearly independent.
Hence

$$m = \dim S \leq \dim \Pi_-(X) = k.$$

This concludes the proof. □

Remark 4.1. As shown in 4.1, for the Taylor–Green case $\mathcal{H}e_{j,k} = \mu_{j,k}\, e_{j,k}$. So the sign of $\mathcal{H}$ is only determined by $j^2 + k^2$. In particular, $\mathcal{H}$ has exactly four negative directions $N = \mathrm{span}\{\sin x, \sin y, \cos x, \cos y\}$. By Lemma 4.1, this implies that the Taylor–Green operator can have at most four linearly independent unstable eigenfunctions.

5 Instability criterion for Hamiltonian operators

This section presents the main theoretical result of the original paper: a finite-dimensional instability criterion for Hamiltonian operators. Under a suitable splitting of $\mathcal{H}$, the spectral problem for the infinite-dimensional operator $\mathcal{L}$ is encoded by a finite-dimensional operator $\mathcal{M}_\lambda$. The eigenvalues of $\mathcal{L}$ outside of $i\,\mathbb{R}$, are then characterized as the zeros of the holomorphic function $\Phi(\lambda) = \det(\mathcal{M}_\lambda)$.

The idea is motivated by the previous subsection: unstable directions of $\mathcal{L}$ can only arise from the negative directions of $\mathcal{H}$. In the Taylor–Green case these directions form a finite-dimensional subspace, so it is natural to isolate them from the remaining coercive part of the operator. In this way, all potentially unstable behaviour is confined to a finite-dimensional component, while positivity controls the rest of the space. We now make this precise by stating the abstract criterion, then discussing the simpler rank-one case, and finally collecting some useful properties of $\Phi(\lambda)$.

Theorem 5.1. *Let $\mathcal{L} = \mathcal{J}\mathcal{H}$ be a Hamiltonian operator as in Definition 4.1. Assume that $\mathcal{H}$ admits a decomposition $\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u$ satisfying the following properties:*

1. *$\mathcal{H}_s : X \to X$ is self-adjoint and*
$$\inf\{\langle \mathcal{H}_s f, f\rangle_X : \|f\|_X = 1\} > 0;$$
2. *$\mathcal{H}_u : X \to V_u$ has finite rank, with*
$$n := \dim(V_u) < +\infty;$$
3. *$\mathcal{H}_s$ and $\mathcal{H}_u$ commute.*

Define the skew-adjoint operator $\mathcal{T} := \mathcal{H}_s^{1/2}\mathcal{J}\mathcal{H}_s^{1/2}$. For every $\lambda \in \mathbb{C} \setminus i\mathbb{R}$, let $\mathcal{M}_\lambda : V_u \to V_u$ be the finite-dimensional operator and $\Phi : \mathbb{C} \setminus i\mathbb{R} \to \mathbb{C}$ be given by

$$\mathcal{M}_\lambda := \mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T}, \qquad \Phi(\lambda) := \det(\mathcal{M}_\lambda). \tag{5.1}$$

Then $\ker(\mathcal{L} - \lambda)$ and $\ker(\mathcal{M}_\lambda)$ are isomorphic. In particular,

$$\lambda \in \sigma(\mathcal{L}) \setminus i\mathbb{R} \quad \Longleftrightarrow \quad \Phi(\lambda) = 0. \tag{5.2}$$

Remark 5.1. In the special case $n = 1$, the function Φ takes a more explicit form. Indeed, let $V_u = \mathrm{span}\{E_u\}$ and choose $h_u \in \mathbb{R}$, $s_u > 0$ such that

$$\mathcal{H}_u(E_u) = -h_u E_u, \qquad \mathcal{H}_s(E_u) = s_u E_u.$$

Then one obtains

$$\Phi(\lambda) = 1 - \frac{h_u}{s_u\|E_u\|_X^2}\left\langle (\mathcal{T} - \lambda)^{-1}\mathcal{T}E_u, E_u\right\rangle_X. \tag{5.3}$$

Proof. Let $f \in \ker(\mathcal{L} - \lambda)$. Since $\mathcal{L} = \mathcal{J}(\mathcal{H}_s + \mathcal{H}_u)$, we have $(\mathcal{J}(\mathcal{H}_s + \mathcal{H}_u) - \lambda)f = 0$ or equivalently

$$(\mathcal{J}\mathcal{H}_s - \lambda)f = -\mathcal{J}\mathcal{H}_u f. \tag{5.4}$$

Since $\mathcal{H}_s$ is self-adjoint and coercive, it admits a unique positive square root $\mathcal{H}_s^{1/2}$, and this operator commutes with $\mathcal{H}_u$. Applying $\mathcal{H}_s^{1/2}$ to (5.4), doesn't alter its solutions because $\mathcal{H}_s^{1/2}$ is bijective. We therefore obtain an equation which is equivalent to $f \in \ker(\mathcal{L} - \lambda)$

$$(\mathcal{T} - \lambda)(\mathcal{H}_s^{1/2} f) = -\mathcal{T}\mathcal{H}_u(\mathcal{H}_s^{-1/2} f), \tag{5.5}$$

where

$$\mathcal{T} := \mathcal{H}_s^{1/2}\mathcal{J}\mathcal{H}_s^{1/2}.$$

Let us now prove implication ($\Rightarrow$). For $\lambda \in \mathbb{C} \setminus i\mathbb{R}$, it holds $\mathrm{Re}(\lambda) \neq 0$ and $\mathcal{T}$ is skew-adjoint,therefore the operator $\mathcal{T} - \lambda$ is invertible. This can be seen immediately by recalling the definition of spectrum and resolvent set and of an operator: $\sigma(A) := \mathbb{C}\backslash\rho(A)$, $\rho(A) := \left\{\lambda \in \mathbb{C} : \ A - \lambda I : D(A) \to X \text{ is bijective and } (A - \lambda I)^{-1} \in \mathcal{L}(X)\right\}$. and the fact that for skew-adjoint operators like $\mathcal{T}$, one has $\sigma(\mathcal{T}) \subset i\mathbb{R}$.
Now apply $(\mathcal{T} - \lambda)^{-1}$ to both sides of (5.5), and then compose it with $\mathcal{H}_u\mathcal{H}_s^{-1}$, we get

$$\mathcal{H}_u\mathcal{H}_s^{-1/2} f + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T}(\mathcal{H}_u\mathcal{H}_s^{-1/2} f) = 0. \tag{5.6}$$

Thus, if $f \in \ker(\mathcal{L} - \lambda)$, $\lambda \in \mathbb{C} \setminus i\mathbb{R}$ then $z := \mathcal{H}_u \mathcal{H}_s^{-1/2} f$ belongs to $\ker(\mathcal{M}_\lambda)$. This concludes the proof for the implication ($\Rightarrow$) in relation (5.2)

Conversely, we now need to prove implication ($\Leftarrow$), namely that for every zero of the stability determinant $\Phi(\lambda)$, there exist an eigenvalue of the linear operator $\mathcal{L}$. Let $z \in \ker(\mathcal{M}_\lambda)$, that is,

$$\mathrm{Id}\, z + \mathcal{H}_u \mathcal{H}_s^{-1} (\mathcal{T} - \lambda)^{-1} \mathcal{T} z = 0. \tag{5.7}$$

Define $f := -\mathcal{H}_s^{-1/2} (\mathcal{T} - \lambda)^{-1} \mathcal{T} z$. We claim that $f \in \ker(\mathcal{L} - \lambda)$, this is equivalent to proving that (5.5) holds. By definition of f, the l.h.s in (5.5) can be written as

$$(\mathcal{T} - \lambda)(\mathcal{H}_s^{1/2} f) = -\mathcal{T} z,$$

applying $\mathcal{T}\mathcal{H}_u\mathcal{H}_s^{-1}$ to the definition of f, we obtain a reformulation of the l.h.s. of (5.5)

$$\mathcal{T}\mathcal{H}_u(\mathcal{H}_s^{-1/2} f) = -\mathcal{T}\mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T} z.$$

Now we can use the last two relations to rewrite eq. (5.5) in terms of z

$$\begin{aligned}(\mathcal{T} - \lambda)(\mathcal{H}_s^{1/2} f) + \mathcal{T}\mathcal{H}_u(\mathcal{H}_s^{-1/2} f) &= -\mathcal{T} z - \mathcal{T}\mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T} z \\ &= -\mathcal{T}\Big(z + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T} z\Big) = 0,\end{aligned}$$

where the last step is guaranteed by (5.7). This shows that when z is a zero of the stability determinant, f satisfies the transformed eigenvalue equation, hence $f \in \ker(\mathcal{L} - \lambda)$.

We have therefore constructed two linear maps

$$\begin{aligned}\ker(\mathcal{L} - \lambda) &\longrightarrow \ker(\mathcal{M}_\lambda), \qquad f \longmapsto \mathcal{H}_u\mathcal{H}_s^{-1/2} f, \quad \text{and} \\ \ker(\mathcal{M}_\lambda) &\longrightarrow \ker(\mathcal{L} - \lambda), \qquad z \longmapsto -\mathcal{H}_s^{-1/2}(\mathcal{T} - \lambda)^{-1}\mathcal{T} z,\end{aligned}$$

and these two maps are inverse to each other. Hence

$$\ker(\mathcal{L} - \lambda) \simeq \ker(\mathcal{M}_\lambda).$$

In particular, $\ker(\mathcal{L} - \lambda) \neq \{0\}$ if and only if $\ker(\mathcal{M}_\lambda) \neq \{0\}$, which is equivalent to $\det(\mathcal{M}_\lambda) = 0$. This proves our criterion. □

Let us now provide a proof for the remark 5.1 giving an explicit device to calculate the stability determinant $\Phi(\lambda)$ when $\dim(V_u) = 1$.

Proof. Assume that $V_u = \mathrm{span}\{E_u\}$. Since $M_\lambda : V_u \to V_u$ acts on a one-dimensional space, its determinant is just a scalar. We can compute it by applying M_λ to E_u and projecting back onto E_u, namely

$$\Phi(\lambda) = \det(M_\lambda) = \frac{\langle M_\lambda E_u, E_u\rangle_X}{\|E_u\|_X^2}. \tag{5.8}$$

By definition, $M_\lambda = \mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T}$.Moreover, since V_u is one-dimensional and $\mathcal{H}_u(E_u) = -h_u E_u$, the operator $\mathcal{H}_u$ can be written as

$$\mathcal{H}_u f = -h_u(E_u \otimes E_u) f = -h_u \frac{\langle f, E_u\rangle_X}{\|E_u\|_X^2} E_u.$$

Since $\mathcal{H}_u$ and $\mathcal{H}_s$ commute, we have $\mathcal{H}_u\mathcal{H}_s E_u = \mathcal{H}_s\mathcal{H}_u E_u = -h_u\mathcal{H}_s E_u$.
Thus $\mathcal{H}_s E_u$ is again an eigenvector of $\mathcal{H}_u$ associated with the eigenvalue $-h_u$. Since this eigenspace is precisely $V_u = \mathrm{span}\{E_u\}$, there exists $s_u \in \mathbb{R}$ such that $\mathcal{H}_s E_u = s_u E_u$. Moreover, since $\mathcal{H}_s$ is positive definite, one has $s_u > 0$, and therefore

$$\mathcal{H}_s^{-1} E_u = \frac{1}{s_u} E_u.$$

We now apply these identities to (5.8). Setting $A_\lambda := (\mathcal{T} - \lambda)^{-1}\mathcal{T}$,we obtain

$$
\begin{aligned}
\det(M_\lambda) &= \frac{\langle M_\lambda E_u, E_u\rangle_X}{\|E_u\|_X^2} \\
&= \frac{1}{\|E_u\|_X^2}\left\langle \left[\mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}A_\lambda\right] E_u, E_u\right\rangle_X \\
&= 1 + \frac{1}{\|E_u\|_X^2}\left\langle \mathcal{H}_u\mathcal{H}_s^{-1}A_\lambda E_u, E_u\right\rangle_X \\
&= 1 + \frac{1}{\|E_u\|_X^2}\left\langle \mathcal{H}_s^{-1}A_\lambda E_u, \mathcal{H}_u E_u\right\rangle_X \\
&= 1 - \frac{h_u}{\|E_u\|_X^2}\left\langle \mathcal{H}_s^{-1}A_\lambda E_u, E_u\right\rangle_X \\
&= 1 - \frac{h_u}{\|E_u\|_X^2}\left\langle A_\lambda E_u, \mathcal{H}_s^{-1} E_u\right\rangle_X \\
&= 1 - \frac{h_u}{s_u\|E_u\|_X^2}\left\langle (\mathcal{T} - \lambda)^{-1}\mathcal{T} E_u, E_u\right\rangle_X .
\end{aligned}
$$

□

5.1 Splitting for the Taylor–Green Hamiltonian operator. To better understand theorem 5.1, it is useful to give a concrete picture of the splitting in the Taylor–Green case. To do so we will use a notation similar to that introduced previously but it should only be intended in a self contained way for this portion, not as general definition of the paper. In our setting, introduced in Subsection 2.1, obtaining the required splitting is particularly simple because $\mathcal{H}$ is diagonal and the decomposition acts only on its Fourier multipliers, as shown in section 4.1.

We denote by $I_N := \{(j,k) \neq (0,0) : \mu_{j,k} \leq 0\}$ the set of indices corresponding to the negative and kernel Fourier modes. The splitting $\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u$ is obtained by modifying only the coefficients with $(j,k) \in I_N$. Namely, choose numbers $s_{j,k}, h_{j,k} > 0$ such that

$$
\mu_{j,k} = s_{j,k} - h_{j,k}, \qquad (j,k) \in I_N,
$$

and define

$$
\mathcal{H}_u := -\sum_{(j,k)\in I_N} h_{j,k}\, e_{j,k} \otimes e_{j,k},
$$

$$
\mathcal{H}_s := \sum_{(j,k)\in I_N} s_{j,k}\, e_{j,k} \otimes e_{j,k} + \sum_{(j,k)\notin I_N} \mu_{j,k}\, e_{j,k} \otimes e_{j,k}.
$$

So the Fourier basis is left unchanged: the splitting only redistributes the negative coefficients $\mu_{j,k}$, isolating the *unstable* part in the finite-rank operator $\mathcal{H}_u$ and leaving a *stable* positive operator $\mathcal{H}_s$. It is equally immediate to verify that $\mathcal{H}_u$ commutes with $\mathcal{H}_s$, indeed they are both diagonal in the same basis.
Moreover, the coercivity constant of $\mathcal{H}_s$ is read directly from its Fourier multipliers. It is the minimum between the positive coefficients assigned to the modified finite-dimensional modes and the smallest positive multiplier already present in the stable part. In particular, if $s_{j,k} = s_u \ \forall (j,k) \in I_N$, then in the full space we have

$$
c(\mathcal{H}_s) = \inf_{\|f\|_{L^2}=1} \langle \mathcal{H}_s f, f\rangle_{L^2} = \min\left\{s_u, \frac{1}{2}\right\}.
$$

In this setting, also the fractional powers of $\mathcal{H}_s$ take a particularly simple form. In general, since $\mathcal{H}_s$ is self-adjoint and coercive, the operators $\mathcal{H}_s^{1/2}$ and $\mathcal{H}_s^{-1/2}$ are well defined by the spectral theorem. Here, however, no abstract construction is needed. They are obtained as,

$$
\mathcal{H}_s^{1/2} = \sum_{(j,k)\in I_N} \sqrt{s_{j,k}}\, e_{j,k} \otimes e_{j,k} + \sum_{(j,k)\notin I_N} \sqrt{\mu_{j,k}}\, e_{j,k} \otimes e_{j,k}, \quad \mathcal{H}_s^{-1/2} = \sum_{(j,k)\in I_N} \frac{1}{\sqrt{s_{j,k}}}\, e_{j,k} \otimes e_{j,k} + \sum_{(j,k)\notin I_N} \frac{1}{\sqrt{\mu_{j,k}}}\, e_{j,k} \otimes e_{j,k},
$$

all these quantities are well defines since we know that $s_{j,k}$ are strictly positive by definition and $\mu_{j,k},\ (j,k) \notin I_N$ are strictly positive as well.

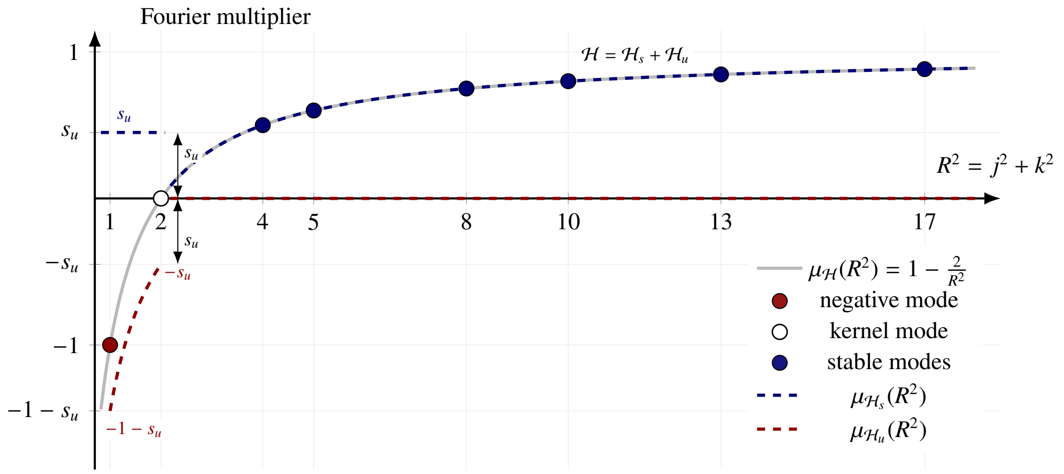

Figure 3: Schematic representation of the Fourier multipliers of the self-adjoint operator $\mathcal{H} = \mathrm{Id} + 2\Delta^{-1}$ as functions of the squared Fourier radius $R^2 = j^2 + k^2$, together with the effect of the splitting $\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u$. The light-gray curve represents $\mu_{\mathcal{H}}(R^2) = 1 - \frac{2}{R^2}$, while the colored dots indicate the admissible discrete radii for $\mathcal{H}$. The negative mode occurs at $R^2 = 1$ and the kernel mode at $R^2 = 2$. The dashed blue graph denotes the multiplier of $\mathcal{H}_s$, whereas the dashed red graph represents $\mu_{\mathcal{H}_u}$: it equals $-1 - s_u$ at the negative mode $R^2 = 1$, equals $-s_u$ at the kernel mode $R^2 = 2$, and vanishes on the stable part.

5.2 Properties of the stability determinant on $\mathbb{R}$. In this subsection we focus on the case of a *real* Hamiltonian operator. By this we mean that the operator commutes with conjugation; equivalently, if f is complex-valued, then

$$\mathcal{L}(\overline{f}) = \overline{\mathcal{L}(f)}.$$

In more intuitive terms, a real operator does not mix the real and imaginary parts of the input.

At this point, Theorem 5.1 has reduced the search for unstable eigenvalues of the Hamiltonian operator $\mathcal{L}$ to the study of the holomorphic function $\Phi(\lambda) := \det(\mathcal{M}_\lambda)$, usually called the *stability determinant*. Since Φ is holomorphic on $\mathbb{C} \setminus i\mathbb{R}$, it is in particular continuous there. Moreover, if $\lambda \in \mathbb{R}$ and $\mathcal{L}$ is real, then $\mathcal{T}$ is also real and $\Phi(\lambda) \in \mathbb{R}$.

This makes the positive real axis especially relevant. Indeed, if Φ changes sign on $(0, +\infty)$, then by continuity it must vanish at some positive real point, and by Theorem 5.1 this gives a real unstable eigenvalue of $\mathcal{L}$.

A first estimate comes from the skew-adjointness of $\mathcal{T}$. From this property we know that its spectrum is contained in $i\mathbb{R}$, therefore the operator $(\mathcal{T} - \lambda)$ is invertible for any $\lambda \in \mathbb{C} \setminus i\mathbb{R}$.

Moreover, for every $u \in D(\mathcal{T})$ one has $\mathrm{Re}\langle(\mathcal{T} - \lambda)u, u\rangle = -\mathrm{Re}(\lambda)\|u\|^2$, since $\mathrm{Re}\langle\mathcal{T}u, u\rangle = 0$. By Cauchy–Schwarz, this gives $|\mathrm{Re}(\lambda)|\,\|u\|^2 \le |\langle(\mathcal{T} - \lambda)u, u\rangle| \le \|(\mathcal{T} - \lambda)u\|\,\|u\|$, hence $|\mathrm{Re}(\lambda)|\,\|u\| \le \|(\mathcal{T} - \lambda)u\|$.

Applying this inequality to $u = (\mathcal{T} - \lambda)^{-1} f$, gives the following result

For every $\lambda \in \mathbb{C} \setminus i\mathbb{R}$ one has

$$\|(\mathcal{T} - \lambda)^{-1}\|_{X \to X} \le \frac{1}{|\mathrm{Re}(\lambda)|}, \tag{5.9}$$

and therefore

$$\lim_{|\lambda| \to +\infty} \Phi(\lambda) = 1. \tag{5.10}$$

Thus, in order to detect zeros of Φ on the positive real axis, it becomes natural to study the behaviour of $\Phi(\lambda)$ as $\lambda \to 0^+$. If the limit exists and is negative, then continuity immediately implies the existence of at least one positive real zero of Φ.

The key point is that the limit of $\mathcal{M}_\lambda$ at the origin is always well defined. To prove this, we first need the following general fact on skew-adjoint operators.

Lemma 5.1 (Strong convergence of the resolvent projection)**.** *Let $\mathcal{T}$ be a closed, densely defined, skew-adjoint operator on a Hilbert space X, and let $P_{\mathcal{T}}$ denote the orthogonal projection onto* $\ker(\mathcal{T})$*. Then the operator family converges strongly as $\lambda \to 0^+$*

$$\lim_{\lambda \to 0^+} (\mathcal{T} - \lambda)^{-1}\mathcal{T} = \mathrm{Id} - P_{\mathcal{T}} \qquad . \tag{5.11}$$

For the proof of this lemma, we refer to Lemma 2.4, p. 9, in the original paper by Cao-Labora, Colombo, Dolce and Ventura [3, Lemma 2.4, p. 9].

We now apply this result to the reduced matrix $\mathcal{M}_\lambda$.

Proposition 5.1 (Limit at zero of $\mathcal{M}_\lambda$)**.** *Under the assumptions of Theorem 5.1, the finite-rank operator $\mathcal{M}_\lambda$ admits the limit*

$$\lim_{\lambda\to 0^+} \mathcal{M}_\lambda = \mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathrm{Id} - P_{\mathcal{T}}) =: \mathcal{M}_0, \tag{5.12}$$

where $P_{\mathcal{T}}$ is the orthogonal projection onto $\ker(\mathcal{T})$*, with $\mathcal{T} := \mathcal{H}_s^{1/2}\mathcal{J}\mathcal{H}_s^{1/2}$.*
Moreover, if P denotes the orthogonal projection onto $\ker(\mathcal{J})$*, then*

$$P_{\mathcal{T}} = \mathcal{H}_s^{-1/2}\, P\,(P\mathcal{H}_s^{-1}P)^{-1}_{|\ker(\mathcal{J})}\, P\,\mathcal{H}_s^{-1/2}. \tag{5.13}$$

In particular, in the rank-one case one obtains

$$\Phi(0) = 1 + \frac{h_u}{s_u}\left(\frac{1}{s_u\|E_u\|_X^2}\left\langle (P\mathcal{H}_s^{-1}P)^{-1}PE_u, PE_u\right\rangle_X - 1\right). \tag{5.14}$$

Proof. In view of Lemma 5.1, the proof of (5.12) is immediate from the definition

$$\mathcal{M}_\lambda = \mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathcal{T} - \lambda)^{-1}\mathcal{T}.$$

Indeed, passing to the limit strongly as $\lambda \to 0^+$ gives $\mathcal{M}_\lambda \longrightarrow \mathrm{Id} + \mathcal{H}_u\mathcal{H}_s^{-1}(\mathrm{Id} - P_{\mathcal{T}})$, which is what we wanted. This is also consistent with the intuition that the limiting operator is obtained by formally setting $\lambda = 0$, provided that the kernel directions are first removed through the projection.

It therefore remains to identify the projection $P_{\mathcal{T}}$. The key insight is that $\ker(\mathcal{T})$ is just the image through $\mathcal{H}_s^{-1/2}$ of the kernel of $\mathcal{J}$ and vice-versa. Indeed, $\mathcal{T}f = 0 \iff \mathcal{H}_s^{1/2}\mathcal{J}\mathcal{H}_s^{1/2}f = 0 \iff \mathcal{J}(\mathcal{H}_s^{1/2}f) = 0$, hence

$$\ker(\mathcal{T}) = \mathcal{H}_s^{-1/2}\ker(\mathcal{J}), \qquad\qquad \mathcal{H}_s^{1/2}\ker(\mathcal{T}) = \ker(\mathcal{J}). \tag{5.15}$$

Now let us construct the orthogonal projection onto this space. Take any $f \in X$ and look for $P_{\mathcal{T}}f$ in the form $P_{\mathcal{T}}f = \mathcal{H}_s^{-1/2}g,\ g \in \ker(\mathcal{J})$. Since $P_{\mathcal{T}}f$ must be the orthogonal projection of f onto $\ker(\mathcal{T})$, the error $f - \mathcal{H}_s^{-1/2}g$ must be orthogonal to every vector of the form $\mathcal{H}_s^{-1/2}h$ with $h \in \ker(\mathcal{J})$. Thus

$$\left\langle f - \mathcal{H}_s^{-1/2}g, \mathcal{H}_s^{-1/2}h\right\rangle_X = 0 \qquad \forall h \in \ker(\mathcal{J}).$$

which, using self-adjointedness of $\mathcal{H}_s^{-1/2}$, can be rewritten as $\langle \mathcal{H}_s^{-1/2}f, h\rangle_X = \langle \mathcal{H}_s^{-1}g, h\rangle_X \quad \forall h \in \ker(\mathcal{J})$.
By Recalling the definition of P, we can express the same relation as $P\,\mathcal{H}_s^{-1}g = P\,\mathcal{H}_s^{-1/2}f$.
Exploiting $g \in \ker(\mathcal{J})$, we may write $g = Pg$, and therefore

$$(P\mathcal{H}_s^{-1}P)g = P\mathcal{H}_s^{-1/2}f.$$

The operator $P\mathcal{H}_s^{-1}P$ is invertible on $\ker(\mathcal{J})$, so

$$g = (P\mathcal{H}_s^{-1}P)^{-1}_{|\ker(\mathcal{J})}\, P\,\mathcal{H}_s^{-1/2}f.$$

Substituting this back into $P_{\mathcal{T}}f = \mathcal{H}_s^{-1/2}g$, we obtain exactly (5.13).

Finally, formula (5.14) follows by substituting (5.13) into (5.12) and then evaluating the determinant in the rank-one case. □

The previous proposition is especially useful because it turns the question "does Φ have a positive real zero?" into the simpler question "what is the sign of $\Phi(0)$?" , which in the rank-one case can be answered easily by direct computation.

Proposition 5.2 (Monotonicity of the stability determinant)**.** *Assume the hypotheses of Theorem 5.1 with* $\dim(V_u) = 1$ *and $h_u > 0$. Then the restriction of Φ to $\mathbb{R}\setminus\{0\}$ is a real, even function, increasing with $|\lambda|$.*

Remark 5.2 (A lower bound for $\Phi(0)$). To rule out positive real eigenvalues through Proposition 5.2, it is enough to show that $\Phi(0) > 0$. Starting from (5.14), one can estimate the term involving $(P\mathcal{H}_s^{-1}P)^{-1}$ from below by using the coercivity of $\mathcal{H}_s$. More precisely, for every $f \in \ker(\mathcal{J})$ with $\|f\|_X = 1$ one has

$$\left\langle (P\mathcal{H}_s^{-1}P)^{-1}f, f\right\rangle_X \geq \inf_{\|g\|_X=1}\langle \mathcal{H}_s g, g\rangle_X =: c(\mathcal{H}_s) > 0. \tag{5.16}$$

If $V_u = \mathrm{span}\{E_u\}$ and $\mathcal{H}_s|_{V_u} = s_u \mathrm{Id}_{V_u}$, then $c(\mathcal{H}_s) = \min\{c(\mathcal{H}|_{V_u^\perp}), s_u\}$.
Choosing $s_u = c(\mathcal{H}|_{V_u^\perp})$, one obtains the lower bound

$$\Phi(0) \geq 1 + \frac{h_u}{s_u}\left(\frac{\|PE_u\|_X^2}{\|E_u\|_X^2} - 1\right). \tag{5.17}$$

This is exactly the form that will be used later in the Taylor–Green analysis.

For the proof of this proposition, we refer to Proposition 2.6, p. 10, in the original paper by Cao-Labora, Colombo, Dolce and Ventura [3, Proposition 2.6, p. 10].

6 Application to the Taylor–Green vortex

We now apply the general theory developed in the previous sections to the Taylor–Green vortex. Let us recall the shape of our operator and its hamiltonian decomposition

$$\omega_E(x,y) := 2\sin x \sin y, \qquad \psi_E(x,y) := -\sin x \sin y, \qquad u_E := \nabla^\perp \psi_E = \begin{bmatrix} \sin x \cos y \\ -\cos x \sin y \end{bmatrix}$$

$$\mathcal{L}_E(w) = -\{\psi_E, (\mathrm{Id} + 2\Delta^{-1})w\}$$

$$\mathcal{J} = -\{\psi_E, \cdot\,\} \quad \text{and} \quad \mathcal{H} = (\mathrm{Id} + 2\Delta^{-1}).$$

Our goal is to study the unstable spectrum of the linearized operator $\mathcal{L}_E$. The strategy is the following. We first decompose $L^2_0(\mathbb{T}^2)$ into invariant subspaces and study $\mathcal{L}_E$ on each of them separately. This is useful because Lemma (4.1) immediately excludes unstable eigenvalues in every subspace that contains no negative direction of $\mathcal{H}$. We are therefore left with only a few relevant subspaces. In the simplest cases, symmetries of our operator reduce the problem to a rank-one situation, and Remark (5.1) shows that the unstable spectrum is then encoded by the scalar stability determinant $\Phi(\lambda)$. Combined with the monotonicity properties established in Proposition (5.2), this allows us to detect or rule out real real unstable eigenvalues simply by studying the sign of $\Phi(0)$. In spaces where we can establish a relation of the kind: $\lambda \in \mathbb{R} \cup i\mathbb{R}$, this corresponds to a full characterization of the unstable spectrum.

6.1 Spectral symmetries. To get a better understanding of the characteristics of $\mathcal{L}_E$, it is useful to record two elementary symmetries of its spectrum . These symmetries explain why eigenvalues, when they are not purely complex, appear in quartets.

The first symmetry comes from the fact that $\mathcal{L}_E$ is a real operator. Indeed, both ψ_E and Δ^{-1} are real, hence

$$\mathcal{L}_E(\overline{f}) = \overline{\mathcal{L}_E f}.$$

Therefore, given an eigenpair $\mathcal{L}_E f = \lambda f$, we can take conjugate both sides of the equation, $\mathcal{L}_E \overline{f} = \overline{\lambda}\,\overline{f}$.
Thus the spectrum is symmetric with respect to the real axis:

$$\lambda \in \sigma(\mathcal{L}_E) \quad \Longrightarrow \quad \overline{\lambda} \in \sigma(\mathcal{L}_E).$$

The second symmetry comes from the Hamiltonian structure $\mathcal{L}_E = \mathcal{J}\mathcal{H}$, where $\mathcal{J}$ is skew-adjoint and $\mathcal{H}$ is self-adjoint. Formally,

$$\mathcal{L}_E^* = (\mathcal{J}\mathcal{H})^* = \mathcal{H}^*\mathcal{J}^* = -\mathcal{H}\mathcal{J}$$

$$\mathcal{H}\mathcal{L}_E\mathcal{H}^{-1} = \mathcal{H}\mathcal{J}\mathcal{H}\mathcal{H}^{-1} = \mathcal{H}\mathcal{J} = -\mathcal{L}_E^*,$$

and therefore

$$\mathcal{L}_E^* = -\mathcal{H}\mathcal{L}_E\mathcal{H}^{-1}. \tag{6.1}$$

Hence, when $\mathcal{H}$ is invertible, $\mathcal{L}_E^*$ is similar to $-\mathcal{L}_E$.

In the Taylor–Green case, $\mathcal{H}$ is not invertible on the whole space because it has a non-trivial kernel. However, this does not affect the unstable spectrum. Indeed, $\ker(\mathcal{H}) \subset \ker(\mathcal{L}_E)$, so the kernel of $\mathcal{H}$ only contributes to the eigenvalue 0. Since we are interested in eigenvalues with $\mathrm{Re}(\lambda) \neq 0$, and in particular $\lambda \neq 0$, we may split

$$X = \ker(\mathcal{H}) \oplus \ker(\mathcal{H})^\perp$$

and study the reduced operator on $\ker(\mathcal{H})^{\perp}$. On this orthogonal complement, the operator $\mathcal{H}$ is invertible, so the similarity relation (6.1) applies to the reduced operator. Consequently, Whenever the restriction of $\mathcal{H}$ to the invariant subspace under consideration is invertible, the identity $\mathcal{L}^* = -\mathcal{H}\mathcal{L}\mathcal{H}^{-1}$ shows that the spectrum of the restricted operator is symmetric with respect to the imaginary axis.

Combining the two symmetries, if λ is an eigenvalue with $\mathrm{Re}(\lambda) \neq 0$, then all the following eigenvalues appear

$$\{\lambda, \overline{\lambda}, -\lambda, -\overline{\lambda}\}.$$

6.2 Invariant subspaces of $\mathcal{L}_E$. As discussed in Subsection 2.1, we work in the Fourier space or zero averaged functions with integer frequencies $L^2_0(\mathbb{T}^2)$. One possible decomposition is obtained by separating functions according to parity with respect to the variables x and y:

$$L^2_0(\mathbb{T}^2) = \mathrm{Odd}_x\mathrm{Odd}_y \oplus \mathrm{Odd}_x\mathrm{Even}_y \oplus \mathrm{Even}_x\mathrm{Odd}_y \oplus \mathrm{Even}_x\mathrm{Even}_y. \tag{6.2}$$

where we recall that Even_x and Even_y are restricted to zero average functions.
This splitting can be understood by decomposing the Fourier basis as follows

$$\begin{aligned}\forall (j,k) \in \mathbb{Z}^2 \quad e_{j,k}(x,y) &:= e^{i(jx+ky)}\\ &= \cos(jx+ky) + i\sin(jx+ky)\\ &= (\cos(jx)\cos(ky) - \sin(jx)\sin(ky))\\ &\quad + i(\sin(jx)\cos(ky) + \cos(jx)\sin(ky)).\end{aligned}$$

Therefore each Fourier mode belongs to the span of the separated trigonometric products

$$\cos(jx)\cos(ky), \qquad \sin(jx)\sin(ky), \qquad \sin(jx)\cos(ky), \qquad \cos(jx)\sin(ky), \tag{6.3}$$

which are respectively even-even, odd-odd, odd-even, and even-odd. More directly, we can say that each subspace of the form A_xB_y with $A, B \in \{\mathrm{Even}, \mathrm{Odd}\}$ is generated by the basis

$$\{\varsigma_{\mathrm{A}}(kx)\varsigma_{\mathrm{B}}(jy)\}_{(k,j)\in\mathbb{Z}^2} \qquad \text{where } \varsigma_{\mathrm{Even}}(t) := \cos t, \quad \varsigma_{\mathrm{Odd}}(t) := \sin t. \tag{6.4}$$

We can refine (6.2) further. Fix one of the four spaces A_xB_y, $A, B \in \{\mathrm{Even}, \mathrm{Odd}\}$, it can be split into two subspaces according to the parity of the sum $j + k$:

$$A_xB_y = (A_xB_y)^{[\mathrm{even}]} \oplus (A_xB_y)^{[\mathrm{odd}]}, \tag{6.5}$$

This new subspaces are generated by taking the appropriate base for A_xB_y from those defined in (6.4) and attributing its modes to either $(A_xB_y)^{[\mathrm{even}]}$ or $(A_xB_y)^{[\mathrm{Odd}]}$ based on the parity of $j + k$. *e.g.* $\sin(3x)\cos(5y) \in \mathrm{Odd}_x\mathrm{Even}_y^{[\mathrm{even}]}$, $\cos(6y) \in \mathrm{Even}_x\mathrm{Even}_y^{[\mathrm{even}]}$.

We now check that these spaces are invariant under $\mathcal{L}_E = \mathcal{J}\mathcal{H}$. The action of $\mathcal{H}$ is immediate, since $\mathcal{H} = \mathrm{Id} + 2\Delta^{-1}$ is a Fourier multiplier and therefore preserves both parity and the parity of the sum $j + k$.
It remains to study $\mathcal{J} = -\{\psi_E, \cdot\}$. Since $\psi_E = -\sin x \sin y$, we have

$$\mathcal{J}g = \cos x \sin y\, \partial_y g - \sin x \cos y\, \partial_x g. \tag{6.6}$$

From this formula one sees immediately that $\mathcal{J}$ preserves parity in each variable: differentiation in one variable changes parity, but multiplication by sin in the same variable restores it.
To check wether the refined decomposition (6.5) is also invariant, we need to make sure that the parity of $j+k$ is preserved under the action of $\mathcal{J}$. Without loss of generality let us analyse one of the four cases, take

$$g(x,y) = \sin(jx)\cos(ky).$$

$$\begin{aligned}\text{Then} \quad \mathcal{J}g &= \cos x \sin y\, \partial_y(\sin(jx)\cos(ky)) - \sin x \cos y\, \partial_x(\sin(jx)\cos(ky))\\ &= -k\cos x \sin y\, \sin(jx)\sin(ky) - j\sin x\cos y\, \cos(jx)\cos(ky).\\ &= -\frac{k}{16i}(e^{ix}+e^{-ix})(e^{iy}-e^{-iy})(e^{ijx}-e^{-ijx})(e^{iky}-e^{-iky})\\ &\quad -\frac{j}{16i}(e^{ix}-e^{-ix})(e^{iy}+e^{-iy})(e^{ijx}+e^{-ijx})(e^{iky}+e^{-iky}).\end{aligned}$$

Each term in the expansion is therefore of the form

$$e^{i((\pm j\pm 1)x+(\pm k\pm 1)y)}.$$

In particular, after going back to trigonometric functions, we obtain a linear combination of terms of the form

$$\sin((j\pm 1)x)\cos((k\pm 1)y).$$

Hence the action of $\mathcal{J}$ shifts each frequency by exactly ± 1 in both variables. The other three parity cases are treated in the same way. In each of these terms the quantity $(j\pm 1)+(k\pm 1)$ has the same parity as $j+k$. Hence $\mathcal{J}$ preserves the parity of the sum of the frequencies.

We conclude that all the spaces in (6.5) are invariant under $\mathcal{J}$ and under $\mathcal{H}$, and therefore also under the full linearized operator $\mathcal{L}_E$.

6.3 Spectral stabilities of odd perturbations. Lemma (4.1) tells us that the number of unstable eigenvalues is upper bounded by the number of negative directions of $\mathcal{H}$. The same principle applies after restricting $\mathcal{L}_E$ to any invariant subspace, because each restriction is still a Hamiltonian operator in the sense of Definition (4.1).

We can therefore begin by looking at the negative subspace of $\mathcal{H}$. By Remark 4.1, $N=\operatorname{span}\{\sin x,\sin y,\cos x,\cos y\}$. On the other hand, the invariant spaces introduced in Subsection 6.2 are pairwise orthogonal. A direct inspection shows that $\sin x\in(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$, $\sin y\in(\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$, $\cos x,\cos y\in(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$, and therefore none of the four negative directions belongs to $(\mathrm{Odd}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$ or to $(A_xB_y)^{[\mathrm{even}]}$, $A,B\in\{\mathrm{Even},\mathrm{Odd}\}$. By Lemma 4.1, the spectrum of $\mathcal{L}_E$ is therefore purely imaginary on each of them.

At this point, the analysis reduces to the three remaining invariant subspaces:

$$(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]},\qquad (\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]},\qquad (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}.$$

We first study the two spaces of odd perturbations $(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$ and $(\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$. To compare them, introduce the involution

$$\rho: L^2_0(\mathbb{T}^2)\to L^2_0(\mathbb{T}^2),\qquad \rho f(x,y):=f(y,x),\qquad \rho^2=\mathrm{Id}.$$

Since $\psi_E(x,y)=-\sin x\sin y$ is symmetric in x and y, one checks that

$$\rho\mathcal{H}=\mathcal{H}\rho,\qquad \rho\mathcal{J}=-\mathcal{J}\rho.$$

Consequently,

$$\rho\mathcal{L}_E=-\mathcal{L}_E\rho.$$

Moreover, ρ maps $(\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$ onto $(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$. Hence it is enough to analyse one of the two spaces, because if

$$\mathcal{L}_E f=\lambda f\qquad \text{in } (\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]},$$

then

$$\mathcal{L}_E(\rho f)=-\lambda\rho f\qquad \text{in } (\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}.$$

In particular, the absence of unstable eigenvalues in one space implies the same conclusion for the other.

Proposition 6.1. *The spectrum of $\mathcal{L}_E$ restricted to each of the invariant subspaces* $(\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$ *and* $(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$ *is purely imaginary.*

Proof. By the discussion above, it is enough to work on $(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$. This space contains exactly one negative direction of $\mathcal{H}$, namely $\sin x$. Therefore Lemma 4.1 implies that there can be at most one unstable eigenvalue.

In light of subsection 6.1, every non-real eigenvalue comes together with its complex conjugate. Since our space contains only one negative direction of $\mathcal{H}$, this is impossible for unstable eigenvalues: a pair $\lambda,\overline{\lambda}$ with $\mathrm{Re}(\lambda)>0$, $\Im(\lambda)\neq 0$ would already give two linearly independent unstable eigenfunctions, contradicting Lemma 4.1. Thus every unstable eigenvalue in this subspace must be real and

$$\sigma_{(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}}(\mathcal{L}_E)\subset\mathbb{R}\cup i\mathbb{R}.$$

We are therefore reduced to ruling out positive real eigenvalues. To do so, we use the rank-one form of the instability criterion from Remark 5.1. Let

$$E_u := \sin x, \qquad s_u := \frac{3}{5}, \qquad h_u := \frac{8}{5},$$

and define

$$\mathcal{H}_u f := -\frac{h_u}{\|E_u\|_{L^2}^2} \langle f, E_u\rangle_{L^2} E_u, \qquad \mathcal{H}_s := \mathcal{H} - \mathcal{H}_u.$$

With this choice, $\mathcal{H}_s$ is positive and coercive, and $\mathcal{H}_u$ isolates the only negative direction.

Let P denote the orthogonal projection onto $\ker(\mathcal{J})$. By Remark 5.2,

$$\Phi(0) \geq 1 + \frac{h_u}{s_u}\left(\frac{\|PE_u\|_{L^2}^2}{\|E_u\|_{L^2}^2} - 1\right) = \frac{8}{3}\left(\frac{\|PE_u\|_{L^2}^2}{\|E_u\|_{L^2}^2} - \frac{5}{8}\right).$$

It is therefore enough to prove that

$$\|PE_u\|_{L^2}^2 > \frac{5}{8}\|E_u\|_{L^2}^2 = \frac{5}{4}\pi^2.$$

This estimate is obtained in the original paper by averaging along the streamlines of the Taylor–Green flow; see [3, pp. 15–16, proof of Proposition 3.2]. For our purposes, it is enough to quote the final bound

$$\|PE_u\|_{L^2(\mathbb{T}^2)}^2 \geq \frac{16}{3}\pi = \frac{8}{3\pi}\|E_u\|_{L^2(\mathbb{T}^2)}^2.$$

Hence $\frac{\|PE_u\|_{L^2(\mathbb{T}^2)}^2}{\|E_u\|_{L^2(\mathbb{T}^2)}^2} \geq \frac{8}{3\pi} \approx 0.84 > \frac{5}{8}$. Therefore $\Phi(0)$ is strictly positive.

Finally, Proposition 5.2 shows that Φ is increasing on $(0, +\infty)$ and that $\lim_{\lambda\to+\infty}\Phi(\lambda) = 1$. Hence Φ has no positive real zero. Since we already know that any unstable eigenvalue would have to be real, we conclude that no unstable eigenvalue exists. Therefore the spectrum of $\mathcal{L}_E$ on $(\mathrm{Odd}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$ is purely imaginary, and the same is true on $(\mathrm{Even}_x\mathrm{Odd}_y)^{[\mathrm{odd}]}$, by the symmetry induced by ρ. □

7 Instability of the linearized Euler operator

At this stage, only one invariant subspace remains to be analyzed. In the previous section, we showed that the spectrum of $\mathcal{L}_E$ is purely imaginary on all invariant subspaces except $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$.
The reason is always the same: either the subspace contains no negative direction of $\mathcal{H}$, so Lemma 4.1 immediately rules out instabilities, or it contains only one negative direction and the rank-one criterion can be used to exclude real unstable eigenvalues.

We now turn our attention to the only remaining case. Here the situation is genuinely different: the subspace $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$ contains two negative directions of $\mathcal{H}$, namely $\cos(x)$ and $\cos(y)$, this forces us to use a different argument from that adopted in odd perturbations. The main idea is to once again split this real invariant space into two complex invariant subspaces, so that each of them contains exactly one negative direction. These two spaces are closely related: they are exchanged by the involution ρ, so any spectral information obtained in one of them is automatically inherited by the other. This allows us to reduce the analysis once again to a single invariant space. The next step is to study what an unstable eigenvector would have to look like inside this complex subspace. The idea behind the localization argument is the following. Suppose that f is an eigenvector associated with an eigenvalue λ. Since the only negative direction of $\mathcal{H}$ in the chosen complex sector is generated by $E_{1,0}$, we decompose $f = E_{1,0} + f^{\perp}$,. The Hamiltonian identity $\langle \mathcal{H}f, f\rangle = 0$ then gives bound on the orthogonal part $f^{\perp}$, because $\mathcal{H}$ is coercive on the complement of the negative direction. Using this bound and some calculation, one obtains an explicit estimate for λ. This shows that any unstable eigenvalue must lie in a very small region of the complex plane with strictly positive real part. The remaining eigenvalues are then recovered from the symmetries of $\mathcal{L}_E$.

The final outcome will be that $\mathcal{L}_E$ has exactly four simple eigenvalues outside the imaginary axis, namely

$$\{\lambda_\star, \overline{\lambda_\star}, -\lambda_\star, -\overline{\lambda_\star}\}.$$

7.1 Complex invariant subspaces. We start from the remaining real invariant space $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$. Define the auxiliary real subspace X by

$$X := \left\{ g \in L^2_0(\mathbb{T}^2;\mathbb{C}) \cap (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]} : g(x,y) = \sum_{j,k\geq 0} g_{j,k}\cos(jx)\cos(ky),\ g_{j,k} = 0 \text{ whenever } j \text{ is even and } k \text{ is odd} \right\}.$$

Its orthogonal complement inside $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$ is then

$$X^\perp := \left\{ g \in L^2_0(\mathbb{T}^2;\mathbb{C}) \cap (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]} : g(x,y) = \sum_{j,k\geq 0} g_{j,k}\cos(jx)\cos(ky),\ g_{j,k} = 0 \text{ whenever } j \text{ is odd and } k \text{ is even} \right\}.$$

Since $\mathcal{H}$ is a Fourier multiplier, it preserves both X and $X^\perp$. On the other hand, $\mathcal{J}$ exchanges them: $\mathcal{J}(X) \subset X^\perp$, $\quad \mathcal{J}(X^\perp) \subset X$. as his action has the effect of shifting both frequencies j and k to adjacent ones $(j\pm 1), (k\pm 1)$.. This was previously shown in 6.2.
The involution $\rho f(x,y) := f(y,x)$ also exchanges X and $X^\perp$. As in (6.6), one has

$$\rho\mathcal{H} = \mathcal{H}\rho, \qquad \rho\mathcal{J} = -\mathcal{J}\rho.$$

This suggests combining X and $X^\perp$ into the following complex spaces:

$$(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_+ := \{\, g + i\rho g \ :\ g \in X \,\} \qquad (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_- := \{\, g - i\rho g \ :\ g \in X \,\} \tag{7.1}$$

A direct computation shows that both spaces are invariant under $\mathcal{H}$, $\mathcal{J}$, and therefore under $\mathcal{L}_E = \mathcal{J}\mathcal{H}$. Moreover, we can write the following orthogonal decomposition in $L^2_0(\mathbb{T})$

$$(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]} = (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_+ \oplus (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_-$$

Conjugation maps the + space to the − space and vice versa. Thus, once the spectrum is understood on one of the two spaces, the other follows automatically.

We now equip $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_+$ with an explicit basis. Define the following functions and index set

$$E_{j,k}(x,y) := \cos(jx)\cos(ky) + i(-1)^j\cos(kx)\cos(jy)\,, \quad (j,k) \in \mathbb{Z}^2. \tag{7.2}$$

$$I := \{(j,k) \in \mathbb{Z}^2 :\ j,k \geq 0,\ j+k \text{ odd},\ k \leq j\}. \tag{7.3}$$

Then $\{E_{j,k}\}_{(j,k)\in I}$ is an orthogonal basis of $(\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}_+$.
Let's use this notation to state a relation that will turn out useful later

$$\mathcal{J}E_{1,0} = \frac{i}{2}E_{1,0} - \frac{1}{2}E_{2,1};$$

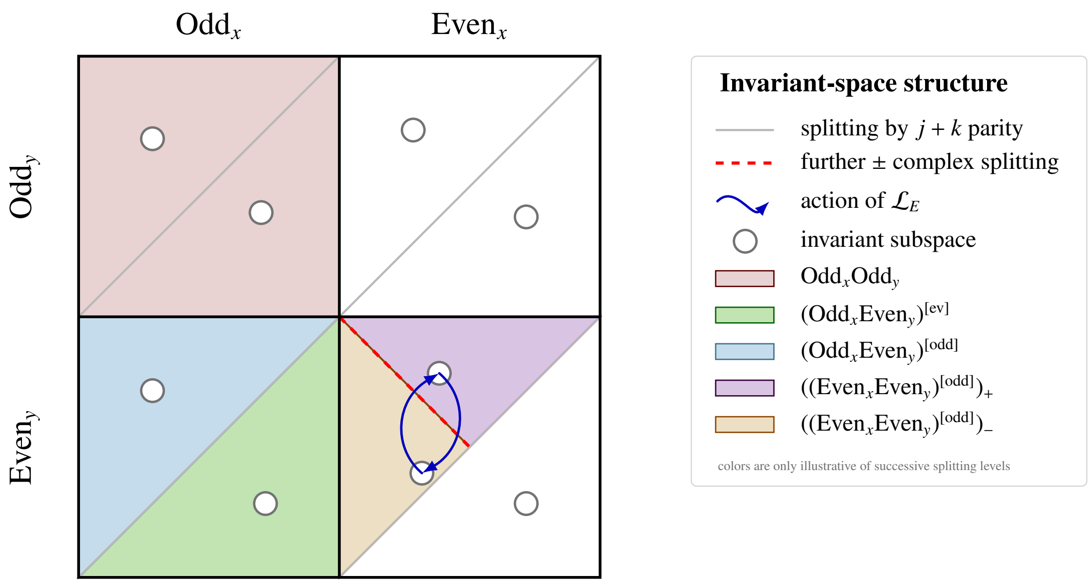

Figure 4: Illustrative scheme of the real invariant subspaces. The decomposition is first organized by parity in x and y, then refined by the parity of $j+k$, and finally by the ± splitting inside a distinguished subspace. The colors are used only exemplarily, to highlight one block for each level of the decomposition.

The next lemma gives a first localization of any unstable eigenvalue in the + space. We quote this result from [3, Lemma 4.4], where a complete proof can be found.

Lemma 7.1. *Let λ be an eigenvalue of $\mathcal{L}_E : (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]} \to (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$ with* $\mathrm{Re}(\lambda) > 0$. *Then it lies in the semicircle*

$$\left\{z \in \mathbb{C} : \ \mathrm{Re}(z) > 0, \ \left|z + \frac{i}{2}\right| \le \frac{3}{10}\sqrt{\frac{5}{6}}\right\}. \tag{7.4}$$

Lemma 7.1 is useful because it provides a specific bounded region were we can search for the unstable eigenvalue. This allows to perform a localized search for the zero of our stability determinant.

Lemma 7.2. *The stability determinant associated with*

$$\mathcal{L}_E : (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]} \to (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$$

has exactly one simple zero in the rectangle

$$[0.10, 0.17] - i[0.57, 0.63].$$

Equivalently, $\mathcal{L}_E$ has one simple unstable eigenvalue in that region.

The proof is computer-assisted and is given in [3, Lemma 4.6 and Section 4.3, pp. 19–22]. It is based on a rigorous winding number argument for the stability determinant. For the purposes of this report, the important point is that Lemma 7.2 rules out the possibility of having only purely imaginary eigenvalues and seals the existence of an unstable eigenvalue in the invariant block under analysis.

7.2 Application of Theorem 1.5. We now explain how the instability criterion is used in this last invariant sector. The construction is exactly the same as in the odd-perturbation case, but now it is carried out inside the complex space

$$(\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}.$$

The unique negative direction of $\mathcal{H}$ in this space is generated by

$$E_u := E_{1,0} = \cos x - i\cos y.$$

We therefore define

$$\mathcal{H}_u f := -\frac{8}{5\|E_u\|_{L^2}^2}\,\langle f, E_u\rangle_{L^2}\,E_u.$$

With this choice,

$$\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u, \qquad s_u = \frac{3}{5}, \qquad h_u = \frac{8}{5},$$

so the assumptions of Theorem 5.1 are satisfied with $n = 1$. The corresponding stability determinant is

$$\Phi(\lambda) = 1 - \frac{8}{3\|E_u\|_{L^2}^2}\left\langle (\mathcal{T} - \lambda)^{-1}\mathcal{T}E_u, E_u\right\rangle_{L^2}, \tag{7.5}$$

where

$$\mathcal{T} := \mathcal{H}_s^{1/2}\mathcal{J}\mathcal{H}_s^{1/2}.$$

Lemma 7.2 tells us that Φ has exactly one simple zero in the rectangle

$$[0.10, 0.17] - i[0.57, 0.63].$$

By Theorem 5.1, this means that

$$\mathcal{L}_E : (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]} \to (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$$

has exactly one simple unstable eigenvalue in that region. We denote it by $\lambda_\star$.

From here, the remaining eigenvalues are recovered by symmetry. Conjugation exchanges the + and − spaces, so it produces the conjugate eigenvalue $\overline{\lambda_\star}$. The Hamiltonian structure then gives symmetry with respect to the imaginary axis, hence also the eigenvalues $-\lambda_\star$ and $-\overline{\lambda_\star}$. This yields the complete quartet outside $i\mathbb{R}$.

We summarize the conclusion in the following proposition, which is the key result of the section.

Proposition 7.1. *The operator*

$$\mathcal{L}_E : (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]} \to (\mathrm{Even}_x\mathrm{Even}_y)^{[\mathrm{odd}]}$$

has exactly one eigenvalue $\lambda_\star$ in the region

$$\lambda_\star \in [0.10, 0.17] - i[0.57, 0.63].$$

Moreover,

$$\sigma(\mathcal{L}_E) \setminus i\mathbb{R} = \{\lambda_\star, \overline{\lambda_\star}, -\lambda_\star, -\overline{\lambda_\star}\},$$

and all four eigenvalues are simple.

Proof. By Lemma 7.2 and Theorem 5.1, the operator $\mathcal{L}_E : (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]} \to (\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$ has a simple eigenvalue $\lambda_\star$ in the rectangle $[0.10, 0.17] - i[0.57, 0.63]$.
Since this complex sector contains only one negative direction of $\mathcal{H}$, Lemma 4.1 implies that $\lambda_\star$ is the only eigenvalue of $\mathcal{L}_E$ in that space with positive real part.
Now, the conjugation ρ maps $(\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$ onto $(\mathrm{Even}_x\mathrm{Even}_y)_-^{[\mathrm{odd}]}$, and $\mathcal{L}_E$ is real.
Therefore, if $\mathcal{L}_E f = \lambda_\star f$ in $(\mathrm{Even}_x\mathrm{Even}_y)_+^{[\mathrm{odd}]}$, then $\mathcal{L}_E\overline{f} = \overline{\lambda_\star}\,\overline{f}$ in $(\mathrm{Even}_x\mathrm{Even}_y)_-^{[\mathrm{odd}]}$. Hence $\overline{\lambda_\star}$ is also an eigenvalue, and it is simple.
Finally we recover the remaining two eigenvalues by making use of the symmetries presented in 6.1. All four are simple, because $\lambda_\star$ and $\overline{\lambda_\star}$ are simple and the symmetries above preserve algebraic multiplicity. This proves the claim. □

8 A novel criterion for the Rank-1 case

The aim of this subsection is to present a novel contribution formulated as part of the semester project. We will relate the sign of the rank-one stability determinant $\Phi(0)$ to a variational quantity on $\ker\mathcal{J}$ and finally show that, in this rank-1 setting, existence of instabilities is equivalent existence of a function with negative energy in $\ker(\mathcal{J})^\perp$. The only additional tool needed is the following identity related to the Legendre transform.

Lemma 8.1 (Legendre transform of a positive quadratic form)**.** *Let Y be a real Hilbert space and let $\mathcal{A} : Y \to Y$ be bounded, self-adjoint, positive and invertible. Then, for every $g \in Y$,*

$$\langle \mathcal{A}^{-1}g, g\rangle_Y = \sup_{\eta\in Y} \{2\langle g, \eta\rangle_Y - \langle \mathcal{A}\eta, \eta\rangle_Y\}.$$

Proof. For every $\eta \in Y$, $2\langle g, \eta\rangle_Y - \langle \mathcal{A}\eta, \eta\rangle_Y = \langle \mathcal{A}^{-1}g, g\rangle_Y - \langle \mathcal{A}(\eta - \mathcal{A}^{-1}g), \eta - \mathcal{A}^{-1}g\rangle_Y$.
Since $\mathcal{A}$ is positive, the second term is non-positive, and equality is attained for $\eta = \mathcal{A}^{-1}g$. □

Theorem 8.1. *Let $\mathcal{L} = \mathcal{J}\mathcal{H}$ be a Hamiltonian operator as in Definition 4.1, and assume that $\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u$ satisfies the hypotheses of Theorem 5.1 with $\dim(V_u) = 1$ and is also invertible.*
Let E_u span the unstable space V_u, and let $\omega, h_u, s_u > 0$ such that

$$\mathcal{H}E_u = -\omega E_u, \qquad \mathcal{H}_u E_u = -h_u E_u, \qquad \mathcal{H}_s E_u = s_u E_u, \quad \text{with } \omega = h_u - s_u > 0.$$

Define

$$\widetilde{C} := \inf_{\substack{f\in\ker(\mathcal{J}) \\ \langle f, E_u\rangle = \|E_u\|^2}} \langle \mathcal{H}^{-1}f, f\rangle.$$

Then

$$\operatorname{sign}\Phi(0) = -\operatorname{sign}\widetilde{C}.$$

Proof. From the rank-one formula for the stability determinant,

$$\Phi(0) = 1 + \frac{h_u}{s_u}\left(\frac{1}{s_u\|E_u\|^2}\left\langle(\mathcal{P}\mathcal{H}_s^{-1}\mathcal{P})^{-1}\mathcal{P}E_u, \mathcal{P}E_u\right\rangle - 1\right).$$

Denoting $A := \left\langle(\mathcal{P}\mathcal{H}_s^{-1}\mathcal{P})^{-1}\mathcal{P}E_u, \mathcal{P}E_u\right\rangle$, then

$$s_u\Phi(0) = s_u - h_u + \frac{h_u}{s_u\|E_u\|^2}A.$$

By Lemma 8.1, applied on $\ker\mathcal{J}$ to the positive operator $\mathcal{P}\mathcal{H}_s^{-1}\mathcal{P}$, we have $A = \sup_{f\in\ker\mathcal{J}}\left\{2\langle f, E_u\rangle - \langle\mathcal{H}_s^{-1}f, f\rangle\right\}$. For $f \in \ker\mathcal{J}$, set $x := \frac{\langle f,E_u\rangle}{\|E_u\|^2}$, thus $\langle f, E_u\rangle = x\|E_u\|^2$. Moreover, since $\mathcal{H}_s^{-1}E_u = \frac{1}{s_u}E_u$, $\mathcal{H}^{-1}E_u = -\frac{1}{\omega}E_u$, and $\mathcal{H}_s^{-1}$ and $\mathcal{H}^{-1}$ coincide on $E_u^\perp$, we get $\mathcal{H}_s^{-1} = \mathcal{H}^{-1} + \left(\frac{1}{s_u} + \frac{1}{\omega}\right)\frac{\langle\,\cdot\,,E_u\rangle}{\|E_u\|^2}E_u$.
Therefore

$$\langle\mathcal{H}_s^{-1}f, f\rangle = \langle\mathcal{H}^{-1}f, f\rangle + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)x^2.$$

Now we use the definition of $\widetilde{C}$ and homogeneity to obtain,

$$\inf_{\substack{f\in\ker\mathcal{J}\\ \langle f,E_u\rangle = x\|E_u\|^2}} \langle\mathcal{H}^{-1}f, f\rangle = x^2\widetilde{C}, \quad \forall x\in\mathbb{R}.$$

Hence

$$A = \sup_{x\in\mathbb{R}}\left\{2\|E_u\|^2 x - \left[\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)\right]x^2\right\}.$$

Optimizing this one-dimensional quadratic expression yields

$$A = \frac{\|E_u\|^4}{\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)}.$$

Substituting this expression into the formula for $s_u\Phi(0)$, we obtain

$$s_u\Phi(0) = s_u - h_u + \frac{h_u}{s_u\|E_u\|^2}\frac{\|E_u\|^4}{\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)}.$$

Using $h_u = s_u + \omega$, this becomes

$$\begin{aligned} s_u\Phi(0) &= -\omega + \frac{h_u\|E_u\|^2}{s_u\left[\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)\right]} \\ &= \frac{-\omega\left[\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)\right] + \frac{h_u}{s_u}\|E_u\|^2}{\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)} \\ &= \frac{-\omega\widetilde{C}}{\widetilde{C} + \|E_u\|^2\left(\frac{1}{s_u} + \frac{1}{\omega}\right)}. \end{aligned}$$

The denominator is positive, because it is the coefficient of the quadratic term in the Legendre transform associated with the positive operator $\mathcal{P}\mathcal{H}_s^{-1}\mathcal{P}$, furthermore, since $s_u > 0$ and $\omega > 0$, we conclude that

$$\operatorname{sign}\Phi(0) = -\operatorname{sign}\widetilde{C}.$$

□

Lemma 8.2. *With the notation of Theorem 8.1, assume that*

$$\widetilde{C} > 0.$$

Then there exists $f \in (\ker\mathcal{J})^\perp$ such that

$$\langle\mathcal{H}f, f\rangle < 0.$$

Proof. Let $K := \ker\mathcal{J}$. If $E_u \in K^\perp$, the conclusion is immediate: indeed, choosing $f = E_u$, Theorem 8.1 gives

$$\langle\mathcal{H}f, f\rangle = \langle\mathcal{H}E_u, E_u\rangle = -\omega\|E_u\|^2 < 0.$$

We may therefore assume that $E_u \notin K^\perp$. In particular, there exists an element of K which has non-zero scalar product with E_u, so the affine set appearing in the definition of $\widetilde{C}$ is non-empty.

Suppose $\exists\,\psi \in K$ such that the following is true

$$f := E_u - \mathcal{H}^{-1}\psi \in K^\perp. \tag{8.1}$$

Indeed, in that case we can write $E_u = f + \mathcal{H}^{-1}\psi,\ f \in K^\perp,\ \psi \in K$.

Using the self-adjointness of $\mathcal{H}$, we obtain

$$\begin{aligned}\langle \mathcal{H}E_u, E_u\rangle &= \left\langle \mathcal{H}(f+\mathcal{H}^{-1}\psi), f+\mathcal{H}^{-1}\psi\right\rangle \\ &= \langle \mathcal{H}f, f\rangle + \langle f, \psi\rangle + \langle \psi, f\rangle + \langle \mathcal{H}^{-1}\psi, \psi\rangle \\ &= \langle \mathcal{H}f, f\rangle + \langle \mathcal{H}^{-1}\psi, \psi\rangle.\end{aligned}$$

Where the mixed terms vanish because $f \in K^\perp$ and $\psi \in K$.
Now $\widetilde{C} > 0$ implies that the quadratic form

$$B(\varphi, \varphi) := \langle \mathcal{H}^{-1}\varphi, \varphi\rangle, \qquad \varphi \in K,$$

is non-negative on K, as discussed in more detail below; therefore $\langle \mathcal{H}^{-1}\psi, \psi\rangle \geq 0$.
Combining this with $\langle \mathcal{H}E_u, E_u\rangle = -\omega\|E_u\|^2 < 0$, we get from (8)

$$\langle \mathcal{H}f, f\rangle = -\omega\|E_u\|^2 - \langle \mathcal{H}^{-1}\psi, \psi\rangle < 0.$$

Thus any decomposition satisfying (8.1) produces a function $f \in K^\perp = (\ker \mathcal{J})^\perp$ with negative $\mathcal{H}$-energy.

It remains to prove that such a decomposition exists. To this end, we first observe that B is positive on K. Let $\varphi \in K$, $\varphi \neq 0$. If $\langle \varphi, E_u\rangle \neq 0$, then, by rescaling φ, we may use the definition of $\widetilde{C}$ and obtain $\langle \mathcal{H}^{-1}\varphi, \varphi\rangle > 0$. If instead $\langle \varphi, E_u\rangle = 0$, then $\varphi \in E_u^\perp$. Since E_u is the only negative direction of $\mathcal{H}$, it is also the only negative direction of $\mathcal{H}^{-1}$. Hence $\mathcal{H}^{-1}$ is positive on $E_u^\perp$, and again $\langle \mathcal{H}^{-1}\varphi, \varphi\rangle > 0$.
Thus B is positive on K.
Now, since $B - \mathrm{Id}$ is compact and B is positive and self-adjoint, the restriction of B to K is a coercive scalar product. Therefore, by the Lax–Milgram theorem, there exists a unique $\psi \in K$ such that

$$\langle \mathcal{H}^{-1}\psi, \eta\rangle = \langle \psi, \mathcal{H}^{-1}\eta\rangle = \langle E_u, \eta\rangle \qquad \forall\, \eta \in K. \tag{8.2}$$

Define

$$f := E_u - \mathcal{H}^{-1}\psi.$$

Then, for every $\eta \in K$, (8.2) gives $\langle f, \eta\rangle = \langle E_u, \eta\rangle - \langle \mathcal{H}^{-1}\psi, \eta\rangle = 0$.
Hence $f \in K^\perp$, so the decomposition required in (8.1) is valid and the proof is complete. □

Lemma 8.3. *With the notation of Theorem 8.1, assume that $\langle \mathcal{H}f, f\rangle < 0$ for some $f \in (\ker \mathcal{J})^\perp$. Then:*

$$\widetilde{C} > 0.$$

Proof. We work by contradiction, assuming $\widetilde{C} \leq 0$. As discussed in the beginning of the proof of Lemma 8.2, the set in the definition of $\widetilde{C}$ is non-empty. Combining our assumption on f with $\widetilde{C} \leq 0$ (arguing by contradiction), there exist $f \in (\ker \mathcal{J})^\perp$ and $g \in \ker \mathcal{J}$ such that

$$\langle \mathcal{H}f, f\rangle < 0, \qquad \langle \mathcal{H}^{-1}g, g\rangle \leq 0, \qquad \langle g, E_u\rangle = \|E_u\|^2.$$

For $\beta \in \mathbb{R}$, set $h_\beta := f + \beta\mathcal{H}^{-1}g$. We evaluate its Krein signature $\langle \mathcal{H}h_\beta, h_\beta\rangle$. Since $\langle f, g\rangle = 0$, the cross term vanishes and

$$\langle \mathcal{H}h_\beta, h_\beta\rangle = \langle \mathcal{H}f, f\rangle + \beta^2\langle \mathcal{H}^{-1}g, g\rangle < 0 \qquad \text{for every } \beta \in \mathbb{R}. \tag{8.3}$$

Also, we choose β such that

$$0 = \langle h_\beta, E_u\rangle = \langle f, E_u\rangle + \beta\langle \mathcal{H}^{-1}g, E_u\rangle \Leftrightarrow \beta = \frac{-\langle f, E_u\rangle}{\|E_u\|^2/\omega},$$

where we used that $\langle \mathcal{H}^{-1}g, E_u\rangle = \langle g, \mathcal{H}^{-1}E_u\rangle = \frac{1}{\omega}\|E_u\|^2$, from the fact that $\langle g, E_u\rangle = \|E_u\|^2$.

With such choice of β, we have

$$\langle \mathcal{H}h_\beta, h_\beta\rangle < 0, \qquad \text{and} \qquad \langle h_\beta, E_u\rangle = 0$$

which is a contradiction since $\mathcal{H}$ is positive on $E_u^\perp$.

□

9 Application to the $(2,2)$ Taylor–Green vortex

In this section we present a novel contribution in the case of the $(2,2)$ Taylor–Green vortex. We formulate a precise criterion for the characterization of the sign of the stability determinant Φ in terms of the amount of energy which is seen by the projection onto $\ker \mathcal{J}_{2,2}$. We then conclude by explicitly constructing a function satisfying the requirements of the criterion, thereby proving the existence of a real spectral instability.

We recall that the $(2,2)$ Taylor–Green vortex is defined by $\psi_{2,2}(x,y) = -\sin(2x)\sin(2y)$, and the corresponding linearized Euler operator has the Hamiltonian form

$$\mathcal{L}_{2,2} = \mathcal{J}_{2,2}\mathcal{H}_{2,2}, \qquad \mathcal{J}_{2,2} := -\{\psi_{2,2}, \cdot\}, \qquad \mathcal{H}_{2,2} := \mathrm{Id} + 8\Delta^{-1}.$$

On the Fourier mode $e^{i(px+qy)}$, the operator $\mathcal{H}_{2,2}$ acts as the multiplier

$$\mathcal{H}_{2,2}\cos(px+qy) = \left(1 - \frac{8}{p^2+q^2}\right)\cos(px+qy).$$

9.1 Transferability of results from the $(1,1)$ operator. We first observe that all the spectral information obtained for the classical $(1,1)$ Taylor–Green vortex are inherited by the new operator in the (2,2) case.
Let $D_2 f(x,y) := f(2x,2y)$ be the dilation operator. It holds

$$\mathcal{J}_{2,2}D_2 = 4D_2\mathcal{J}_{1,1}, \qquad \mathcal{H}_{2,2}D_2 = D_2\mathcal{H}_{1,1}, \qquad \text{and necessarily} \qquad \mathcal{L}_{2,2}D_2 = 4D_2\mathcal{L}_{1,1}.$$

Therefore, if w is an eigenfunction of $\mathcal{L}_{1,1}$ with eigenvalue λ, then $D_2 w$ is an eigenfunction of $\mathcal{L}_{2,2}$ with eigenvalue 4λ:

$$\mathcal{L}_{2,2}D_2 w = 4D_2\mathcal{L}_{1,1}w = 4\lambda D_2 w.$$

In particular, the unstable spectrum found for the $(1,1)$ vortex produces the four rescaled eigenvalues

$$\{4\lambda_\star, 4\overline{\lambda_\star}, -4\lambda_\star, -4\overline{\lambda_\star}\} \subset \sigma(\mathcal{L}_{2,2}) \setminus i\mathbb{R}.$$

For this reason, in the following analysis we will exclude invariant subspaces where the spectrum of $\mathcal{L}_{2,2}$ can be related to that of $\mathcal{L}_E$.

9.2 Invariant subspaces for the $\mathcal{L}_{2,2}$ operator. Similarly to what we did in section 6.2, we will now present the invariant decomposition associated with the $(2,2)$ vortex. Since the Poisson bracket with $\psi_{2,2}$ shifts Fourier frequencies by $(\pm 2, \pm 2)$, the individual parities of the two Fourier frequencies are preserved. Moreover, the checkerboard parity is also preserved. Hence

$$L^2_0(\mathbb{T}^2) = \bigoplus_{\kappa_1,\kappa_2\in\{0,1\}} \bigoplus_{C\in\{0,1\}} L^2_0(\mathbb{T}^2)^{[C]}_{[\kappa_1,\kappa_2]},$$

where $L^2_0(\mathbb{T}^2)^{[C]}_{[\kappa_1,\kappa_2]}$ is generated by Fourier modes of the form

$$e^{i((2j+\kappa_1)x+(2k+\kappa_2)y)}, \qquad j,k\in\mathbb{Z}, \quad \text{with} \quad j+k\equiv C \pmod 2.$$

The operator $\mathcal{H}_{2,2}$ is diagonal on this decomposition, while $\mathcal{J}_{2,2}$ preserves it because it only shifts frequencies by multiples of 2 in both variables. Hence each space $L^2_0(\mathbb{T}^2)^{[C]}_{[\kappa_1,\kappa_2]}$ is invariant under $\mathcal{L}_{2,2}$.

The space $L^2_0(\mathbb{T}^2)_{[0,0]}$ is precisely the one obtained by dilation from the $(1,1)$ operator, and therefore its spectrum is completely transferred from the previous analysis. Moreover, the spaces with indices $[1,0]$ and $[0,1]$ are conjugated by the involution $\rho f(x,y) := f(y,x)$, so it is enough to study only one of them.
We now explain why, for the remaining classes $L^2_0(\mathbb{T}^2)^{[C]}_{[1,0]}$ and $L^2_0(\mathbb{T}^2)^{[C]}_{[1,1]}$, it is enough to consider only the checkerboard component $C = 0$. Recall that a mode in $L^2_0(\mathbb{T}^2)^{[C]}_{[\kappa_1,\kappa_2]}$ has frequencies $(p,q) = (2j+\kappa_1, 2k+\kappa_2)$, $\quad j+k\equiv C \pmod 2$ and consider the involution

$$\rho_x f(x,y) := f(-x,y).$$

It sends the Fourier mode $e^{i((2j+\kappa_1)x+(2k+\kappa_2)y)}$ into $e^{i(-(2j+\kappa_1)x+(2k+\kappa_2)y)}$. If $\kappa_1 = 1$, then $-(2j+1) = 2(-j-1)+1$. Hence, after applying ρ_x, the new integer representative in the x-frequency is $j' = -j-1$. Therefore

$$-(2j+1) \equiv 2(j+1)+1 \pmod 2.$$

Thus ρ_x exchanges the two checkerboard classes:

$$\rho_x : L^2_0(\mathbb{T}^2)^{[0]}_{[\kappa_1,\kappa_2]} \longleftrightarrow L^2_0(\mathbb{T}^2)^{[1]}_{[\kappa_1,\kappa_2]}, \qquad \text{whenever } \kappa_1 = 1.$$

Moreover, ρ_x conjugates the operator on these two spaces, because $\mathcal{H}_{2,2}$ commutes with ρ_x, while $\mathcal{J}_{2,2}$ changes sign under the same involution in the corresponding way. Therefore the two restrictions have the same spectrum. This is precisely the reduction used in [3, Section 5.2].
Consequently, for the classes $[\kappa_1, \kappa_2] = [1, 0]$ and $[\kappa_1, \kappa_2] = [1, 1]$, it is sufficient to study the component $C = 0$.

Hence the genuinely new invariant subspaces, after the transferred $[0, 0]$ block has been removed, are

$$L^2_0(\mathbb{T}^2)^{[0]}_{[1,0]}, \qquad L^2_0(\mathbb{T}^2)^{[0]}_{[1,1]}.$$

9.3 Spectral instability in the space $L^2_0(\mathbb{T}^2)^{[0]}_{[1,1]}$. We now focus on the second one. Although $L^2_0(\mathbb{T}^2)^{[0]}_{[1,1]}$ does not split according to the usual parity in x and y separately, it is invariant under the radial reflection

$$\mathcal{R}f(x, y) := f(-x, -y).$$

Indeed, if $e^{i(jx+ky)}$ belongs to $L^2_0(\mathbb{T}^2)^{[0]}_{[1,1]}$, then so does $e^{-i(jx+ky)}$. We can therefore split this space into its radially even and radially odd parts:

$$L^2_0(\mathbb{T}^2)^{[0]}_{[1,1]} = L^2_0(\mathbb{T}^2)^{[0],c}_{[1,1]} \oplus L^2_0(\mathbb{T}^2)^{[0],s}_{[1,1]}.$$

Following the notation of [3, Section 5.2, p.26], the cosine component has basis

$$\{\cos((2j + 1)x + (2k + 1)y) : j, k \in \mathbb{Z}, \ j + k \equiv 0 \pmod 2, \ j \geq 0\},$$

whereas the sine component has basis

$$\{\sin((2j + 1)x + (2k + 1)y) : j, k \in \mathbb{Z}, \ j + k \equiv 0 \pmod 2, \ j \geq 0\}.$$

The condition $j \geq 0$ is only a choice of representatives, since the modes with indices (j, k) and $(-j-1, -k-1)$ generate the same cosine direction up to sign.
The two spaces $L^2_0(\mathbb{T}^2)^{[0],c}_{[1,1]}$ and $L^2_0(\mathbb{T}^2)^{[0],s}_{[1,1]}$ are again spectrally equivalent. Indeed, the translation-reflection symmetry

$$\rho_t f(x, y) := f\left(\frac{\pi}{4} - x, \frac{\pi}{4} - y\right)$$

maps the cosine basis into the sine basis, up to signs, and conjugates the two restrictions of $\mathcal{L}_{2,2}$.
Hence it is enough to work on the invariant block

$$\mathcal{X} := L^2_0(\mathbb{T}^2)^{[0],c}_{[1,1]}.$$

As discussed in previous sections, the linearized operator has the Hamiltonian form $\mathcal{L} = \mathcal{J}\mathcal{H}$. On the space $\mathcal{X}$, the operator $\mathcal{H} = \mathcal{H}_{2,2} = \mathrm{Id} + 8\Delta^{-1}$ has exactly one negative direction, namely

$$E_u(x, y) := \cos(x + y), \qquad \mathcal{H}E_u = \left(1 - \frac{8}{1^2 + 1^2}\right)E_u = -3E_u.$$

We now present the instability argument in this new block $\mathcal{X} := L^2_0(\mathbb{T}^2)^{[0],c}_{[1,1]}$. On this space we consider the restricted Hamiltonian operator $\mathcal{L}_{2,2}|_{\mathcal{X}_c} = \mathcal{J}_{2,2}\mathcal{H}_{2,2}|_{\mathcal{X}_c}$, $\quad \mathcal{J}_{2,2} := -\{\psi_{2,2}, \cdot\}$, $\quad \mathcal{H}_{2,2} := \mathrm{Id} + 8\Delta^{-1}$. As discussed above, the space $\mathcal{X}$ is invariant under both $\mathcal{J}_{2,2}$ and $\mathcal{H}_{2,2}$. Furthermore, $\mathcal{H}_{2,2}$ is invertible on $\mathcal{X}$, because the equation $p^2 + q^2 = 8$ cannot hold with p, q both odd.
We can therefore apply the rank-one criterion of Theorem 8.1. In this setting. By Lemma 8.3, it is enough to find a function

$$f \in (\ker \mathcal{J}_{2,2})^\perp \cap \mathcal{X} \qquad \text{such that} \qquad \langle \mathcal{H}_{2,2} f, f\rangle_{L^2} < 0.$$

The useful observation is that every function in the range of $\mathcal{J}_{2,2}$ is automatically orthogonal to $\ker \mathcal{J}_{2,2}$. Indeed, using the skew-adjointness of $\mathcal{J}_{2,2}$, $\quad \langle \mathcal{J}_{2,2} g, h\rangle_{L^2} = -\langle g, \mathcal{J}_{2,2} h\rangle_{L^2} = 0 \quad \forall h \in \ker \mathcal{J}_{2,2}$.
Thus

$$\mathrm{Ran}(\mathcal{J}_{2,2}) \subset (\ker \mathcal{J}_{2,2})^\perp.$$

It remains only to find a trigonometric polynomial $g \in \mathcal{X}$ such that $\mathcal{J}_{2,2}\, g$ has negative $\mathcal{H}_{2,2}$-energy.

We claim that the following choice works:

$$g(x, y) := \cos(3x - y) - \cos(x - 3y).$$

Both Fourier modes belong to $\mathcal{X}$: for $\cos(3x - y)$ the frequencies are $(3, -1) = (2 \cdot 1 + 1, 2 \cdot (-1) + 1)$, hence $(j, k) = (1, -1)$; for $\cos(x - 3y)$ they are $(1, -3) = (2 \cdot 0 + 1, 2 \cdot (-2) + 1)$ and therefore $(j, k) = (0, -2)$. In both cases the indices j, k have even sum, so $g \in \mathcal{X}$.
Let us recall that $\psi_{2,2}(x, y) = -\sin(2x)\sin(2y)$, and $\mathcal{J}_{2,2}u = 2\cos(2x)\sin(2y)\,\partial_y u - 2\sin(2x)\cos(2y)\,\partial_x u$.

For the function g above,

$$\partial_x g = -3\sin(3x - y) + \sin(x - 3y), \qquad \partial_y g = \sin(3x - y) - 3\sin(x - 3y).$$

Using the product-to-sum trigonometric identities, we obtain

$$\begin{aligned} f := \mathcal{J}_{2,2}g = {} & 2\cos(x + y) + 2\cos(3x - y) + 2\cos(x - 3y) \\ & - 2\cos(5x + y) - 2\cos(x + 5y) - \cos(5x - 3y) - \cos(3x - 5y). \end{aligned}$$

It remains to compute its $\mathcal{H}_{2,2}$-energy. The Fourier modes appearing in the expansion of f are mutually orthogonal in $L^2(\mathbb{T}^2)$, and with the choice of scalar product introduced in subsection 2.1,
$\|\cos(px + qy)\|^2_{L^2(\mathbb{T}^2)} = 2\pi^2, \forall (p, q) \neq (0, 0)$. Finally, we know how $\mathcal{H}_{(2,2)}$ acts on each mode. Therefore,

$$\begin{aligned} \langle \mathcal{H}_{2,2}f, f\rangle_{L^2} &= 2\pi^2 \left[4\left(1 - \frac{8}{2}\right) + 2 \cdot 4\left(1 - \frac{8}{10}\right) + 2 \cdot 4\left(1 - \frac{8}{26}\right) + 2\left(1 - \frac{8}{34}\right)\right] \\ &= 2\pi^2 \left[-12 + \frac{8}{5} + \frac{72}{13} + \frac{26}{17}\right] \\ &= -\frac{7364}{1105}\pi^2 < 0. \end{aligned}$$

Thus we have explicitly constructed a function

$$f \in (\ker \mathcal{J}_{2,2})^{\perp} \cap X_c \qquad \text{with} \qquad \langle \mathcal{H}_{2,2}f, f\rangle_{L^2} < 0.$$

By Lemma 8.3, this implies $\widetilde{C} > 0$. Then Theorem 8.1 gives $-\operatorname{sign}\widetilde{C} = \operatorname{sign}\Phi(0) < 0$.
On the other hand, as in Proposition 5.2, the rank-one stability determinant satisfies $\lim_{\lambda\to+\infty}\Phi(\lambda) = 1$.
Hence, by continuity, there exists $\lambda_\triangle > 0$ such that

$$\Phi(\lambda_\triangle) = 0.$$

Theorem 5.1 therefore yields a real unstable eigenvalue of $\mathcal{L}_{2,2}$ in the block $\mathcal{X}$.
Finally, since $\mathcal{X}$ contains only one negative direction of $\mathcal{H}_{2,2}$, Lemma 4.1 rules out the presence of two linearly independent unstable eigenfunctions in this block. Moreover, the same argument used in Subsection 6.1 gives the symmetry with respect to the imaginary axis. Therefore the point spectrum of the restricted operator outside the imaginary axis consists of the real pair

$$\{\lambda_\triangle, -\lambda_\triangle\}.$$

By the symmetry equivalences between the cosine and sine components discussed above, the same instability is inherited by the corresponding conjugated blocks of $L^2_0(\mathbb{T}^2)_{[1,1]}$.
We now clarify the multiplicity in this block. The space $\mathcal{X}$ contains exactly one negative direction of $\mathcal{H}_{2,2}$, namely the direction generated by $\cos(x + y)$. Therefore Lemma 4.1 implies that $\mathcal{L}_{2,2}|_{\mathcal{X}}$ can have at most one linearly independent unstable eigenfunction. In particular, the positive real eigenvalue found above is the only eigenvalue of $\mathcal{L}_{2,2}|_{\mathcal{X}}$ with positive real part, and its eigenspace is one-dimensional.
The stable counterpart is recovered from the Hamiltonian symmetry. Indeed, on the block $\mathcal{X}$ the operator $\mathcal{H}_{2,2}$ is invertible, and the same argument used in Subsection 6.1 gives the symmetry of the spectrum with respect to the imaginary axis. Since $\lambda_\triangle$ is real, this symmetry gives the eigenvalue $-\lambda_\triangle < 0$. Thus the restricted operator has, outside the imaginary axis, the real pair

$$\sigma(\mathcal{L}_{2,2}|_{\mathcal{X}}) \setminus i\mathbb{R} = \{\lambda_\triangle, -\lambda_\triangle\}.$$

Here $\lambda_\triangle$ is the unstable eigenvalue and $-\lambda_\triangle$ is the corresponding stable eigenvalue. Both have one-dimensional eigenspace inside the single block $\mathcal{X}$.

Finally, this should be distinguished from the multiplicity of the same eigenvalues in the full space. The block $\mathcal{X}$ is only one of the four symmetry-related components of $L^2_0(\mathbb{T}^2)_{[1,1]}$, namely

$$L^2_0(\mathbb{T}^2)^{[0],c}_{[1,1]}, \qquad L^2_0(\mathbb{T}^2)^{[0],s}_{[1,1]}, \qquad L^2_0(\mathbb{T}^2)^{[1],c}_{[1,1]}, \qquad L^2_0(\mathbb{T}^2)^{[1],s}_{[1,1]}.$$

The symmetry equivalences discussed above map the spectral problem on $\mathcal{X}$ onto the spectral problem on the other three components. Consequently, the same pair

$$\{\lambda_\triangle, -\lambda_\triangle\}$$

appears in each of the four blocks. Therefore, as eigenvalues of the full operator $\mathcal{L}_{2,2} : L^2_0(\mathbb{T}^2) \to L^2_0(\mathbb{T}^2)$, both $\lambda_\triangle$ and $-\lambda_\triangle$ have multiplicity four, while in each individual block they appear with multiplicity one.

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